# Science Tasks Organization

Preliminary notes

Version 001

SS 2025

THz Group
Faculty of Physics
LMU München

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File: `Q:\THz-Office\StudentOrganize\ScienceTasks\AAScienceTasks.tex`

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# Latest changes

1. 02.12.2025, VP: Added Section 10 "Electrical characterisation"
2. 02.12.2025, SB: Added Section 9 "PVD deposition parameters"
3. 02.12.2025, SB: Updated Section 8
4. 20.08.2025, RK: Started "Thickness of $MoO_x$"
5. 18.08.2025, RK: More on "Side injection"
6. 13.08.2025, RK: Started "Lab cleaning"
7. 13.08.2025, RK: Modification "Side injection"
8. 11.08.2025, RK: Started "Side injection"
9. 11.08.2025, RK: Started "How to planarize Teonex"
10. 10.08.2025, RK: Set up of this document

# 1 How to planarize Teonex samples

**Date of issue:** 11.08.2025 **Last Changes:**

- 18.09.2025, SB, VP: recipe for spin coating of PS & parameters for depositing Trip-C12 on PaN+PS
- File name: 20250902

- File name:20250903001

## 1.1 Background

- THz transmission through Teonex is higher than through sapphire.
- Deposition on Teonex substrates is easier (16 at once).
- Many experiments can be performed on Teonex substrates:
  - AFM studies of roughness
  - THz $dS/S$ scans on ms scales
  - THz $dS/S$ in dependence on angle
  - THz $dS/S$ in dependence on illumination
  - THz $dS/S$ in dependence on temperature (slow)

## 1.2 Questions

### 1.2.1 Question substrate size

1. Find spin coating parameters for PS on PaN

   **Answer:** Use 1% wt PS in Toluene, spin at 500 rpm (5 s) + 4000 rpm (50 s), then bake at 90°C for 10 minutes and allow it to cool gradually for 10 minutes. Static dispense method was used to drop the solution on the substrate.
2. Which chuck is best?

   **Answer:** For Teonex, an acrylic chuck with glass is recommended (apply one drop of isopropanol and then place the sample on the glass). For sapphire or glass, a metal chuck works best.
3. What is the best substrate size?

   **Answer:** Recommendation: spin coat on 20 mm × 20 mm substrates and later cut to 10 mm × 10 mm. **Note:** As of 08.09.2025, samples of 20 mm × 20 mm have been prepared and we cut them.
4. Can reasonable results be obtained?

   **Answer:** Yet to be verified.
   - Is the standard solution suitable? **Yes** (1% wt PS in Toluene).

- Are the standard spin-coating parameters suitable? **Yes**, 60–80 nm thickness can be achieved with this recipe.

### 1.2.2 Question surface roughness

1. Take AFM images.

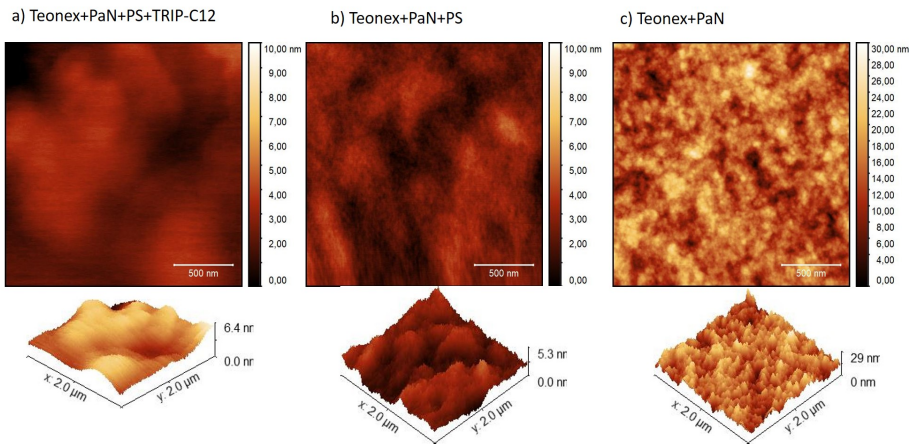



Figure 1.1: (a) Teonex+PaN+PS+TRIP-C12. (b) Teonex+PaN+PS. (c)Teonex+PaN

2. Obtain HHCs

- How much does the surface roughness decrease, see HHC(30nm)?

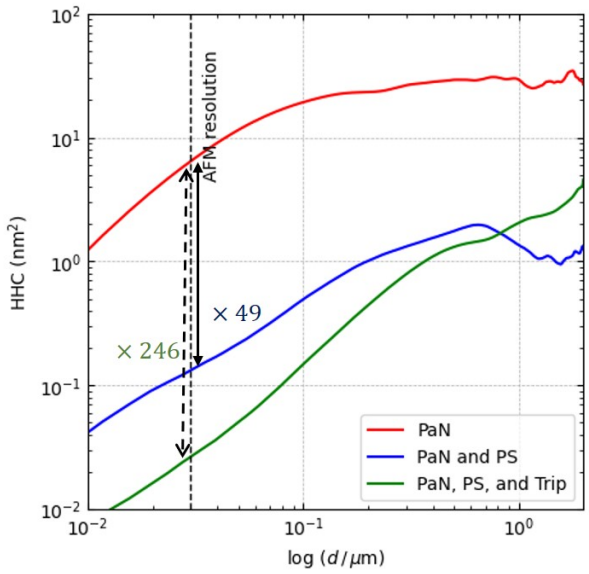



Figure 1.2: Height–Height Correlation Function (HHCF) for evaluating surface roughness.

**Answer:** The surface roughness was significantly reduced upon planarization. With the PS+TRIP-C12 (5 nm) layer, the roughness decreased by approximately a factor of 246, while with PS alone the reduction was more pronounced, improving by nearly a factor of 49.

## 1.3 Results

- Spin-coating of 1% wt PS in Toluene with the optimized parameters produced uniform films with thicknesses in

the range of 60–80 nm.

- AFM and HHCF analysis confirmed a significant reduction in surface roughness after planarization:
  - With PS+TRIP-C12 (5 nm), the roughness decreased by approximately a factor of 246.
  - With PS alone, the roughness decreased by nearly a factor of 49.

## 1.4 Conclusion

- The optimized recipe for PS planarization on Teonex substrates is reliable and reproducible.
- Both PS and PS+TRIP-C12 layers effectively reduce surface roughness, though PS+TRIP-C12 shows superior improvement.

## 1.5 Final recipe

| **Parameter** | **Final Recipe** |
|---|---|
| Solution | 1% wt PS in Toluene |
| Dispense method | Static dispense (drop solution then start) |
| Spin coating | 500 rpm for 5 s, followed by 4000 rpm for 50 s |
| Baking | 90°C for 10 min, then gradual cooling for 10 min |
| Chuck | For Teonex: acrylic chuck with glass (apply 1 drop of isopropanol before placing the Teonex). For sapphire/glass: metal chuck without adherent. |
| Substrate size | Recommended: 20 mm × 20 mm, later cut to 10 mm × 10 mm |
| Expected film thickness | 60–80 nm |

Table 1.1: Final recipe for spin coating of PS

| **Parameter** | **Comments** |
|---|---|
| Weight | 20mg |
| Pressure | $5 \times 10^{-7}$ mbar |
| Substrate temperature | 30°C |
| Temperature (Ramp 1) | 100°C |
| Soak 1 | 2000s |
| Temperature (Ramp 2) | 190°C |
| Soak 2 | 2000s |
| Deposition temperature | Starts after 200°C |
| Thickness | 5nm |

Table 1.2: Parameters for depositing Trip C12 on PaN+PS

# 2 Properties of side injection

**Date of issue/modification:** 13.08.2025

## 2.1 Background

- Until 08.2025, the THz devices had varactor architecture (s. Fig. 2.1(a)).
- More advanced would be side injection devices (s. Fig. 2.1(b)). Their advantages are:
  - Metal atoms cannot contaminate the channel.
  - There is no need for cooling the samples during deposition of the injection contacts.

### 2.1.1 Architectures

Tow possible architectures are shown in Figs. 2.2 and 2.3:

- semiconductor-bottom, injection-top geometry
- injection-bottom, semiconductor-top geometry

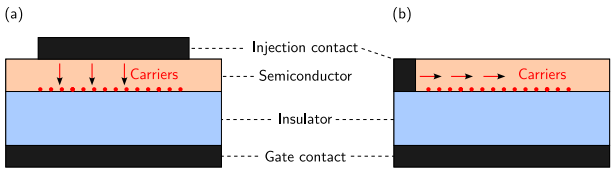



Figure 2.1: (a) Varactor type THz device. (b) Device for side injection of charge carriers.

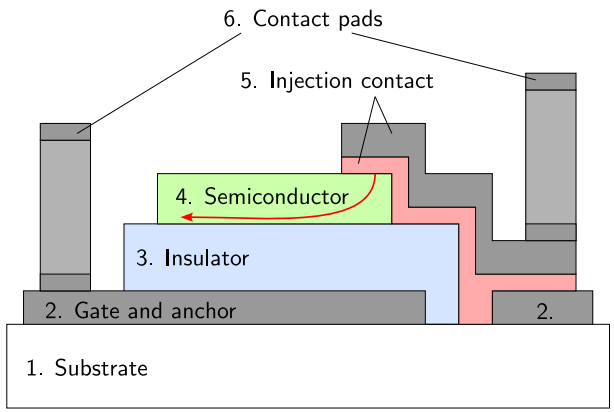



Figure 2.2: Cross section through a device for side injection. The architecture is semiconductor-bottom injection-top geometry. Details are given on next page.

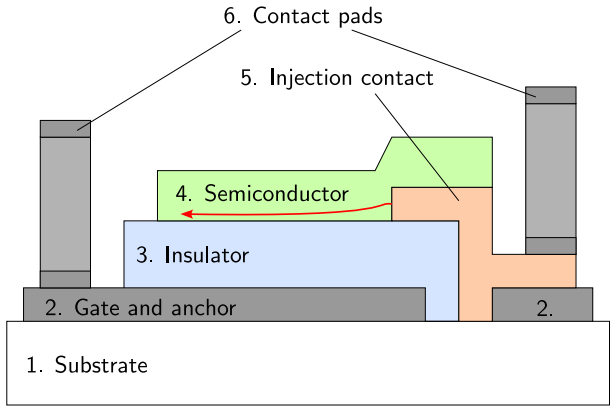



Figure 2.3: Cross section through a device for side injection. The architecture is injection-bottom semiconductor-top geometry. Details are given on next page.

### 2.1.2 Properties of injection contacts

**Semiconductor-bottom injection-top geometry:**

- Fig. 2.2
- alias for geometry: SBIT
- The $MoO_x$-layer should be similar to those used in varactor-geometry. A reasonable thickness of the $MoO_x$-layer may be 1.5 nm.
- The Cr-Layer can be thicker, because the THz radiation propagates in the center of the device. A ticker layer at the injection contact doesn't block THz radiation. A reasonable thickness of the Cr-layer may be 20 nm.

**Injection-bottom semiconductor-top geometry:**

- Fig. 2.3
- alias for geometry: STIB
- In this geometry, an $MoO_x$-layer may be useless.
- Instead of a Cr-layer a Au-layer can be used:
  - Au is stable and ductile, and easy to deposit.
  - Ion migration fo the Au into the channel can be leglected because of the aspect ration of the semiconductor layer.
  - A reasonable thickness of the Au-layer may be 20 nm.

## 2.2 Tasks

**Mind:** Nothing done, so far

### 2.2.1 Device Fabrication

1. Top views of the device architectures are shown in Fig. 2.4.
2. On the same Teonex substrate, the following device types were fabricated:
   - **Varactor-type devices:** 12 devices fabricated on Teonex substrates (20 mm $\times$ 20 mm) in three groups: PaN, PaN+PS, and PaN+PS+TRIP-C12. The semiconductor layer consisted of C8-BTBT-C8. A $MoO_x$ layer (3.5 nm) was deposited under cooling, followed by a 12 nm Cr layer (no wagging).
   - **Devices with side injection (STIB, ITSB):** Fabricated using mask No. 71. Device stack: 7 nm Cr layer + 225 nm PaN, deposited on 16 Teonex substrates (20 mm $\times$ 20 mm).
3. Required masks:
   **available**.
4. As of 19.08.2025: $MoO_x$ layer should not exceed 1.5 nm thickness.

   It is explained in Section 5.

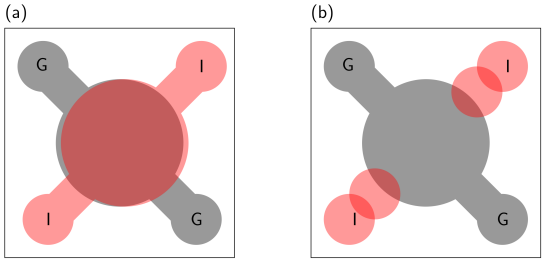



Figure 2.4: Top views on to devices with varactor architecture (a) and with side-injection architecture.

### 2.2.2 Device characterization

1. Obtain for all devices the resistance between the tow gate contacts $R_{GG}$ and between the two injection contacts $R_{II}$.
2. For the varactor type devices measure $C_{GI}$.
3. Mind that the side-injection devices have vanishingly small capacitance $C_{GI}$ as long as no high gate voltage is applied!

| Layer | $R_{GG}$ | $R_{II}$ | $C_{GI}$ |
|---|---|---|---|
| # | kΩ | kΩ | nF |
| PaN | 2.3 | 11 | 2.6 |
| PS | 2.3 | 11.6 | 2.2 |
| TRIP-C12 | 2.1 | - | - |

Table 2.1: Varactor type devices measurements

### 2.2.3 THz characterization

1. Record for all types of devices THz ms-scans.
   - Are the modulations $dS/S$ comparable?
   - Is the speed of the devices comparable? Slow devices will not reproduce rectangles in the ms-scan.

## 2.3 Experimental results

## 2.4 Conclusion

## 2.5 Suggested device structure

# 3 3D sketches of devices

## 3.1 Layout of varactor device

### 3.1.1 Applications

- THz measurements

### 3.1.2 File locations

**CAD file name:** `Varactor001014.FCStd`

**Directory:**
`Q:\THz-Lab\Lab Management\Technical Drawings\Sample3D\..`

**Scalable vector graphics:** `yySec003004001.svg`
**Directory:**
`\..\StudentOrganize\ScienceTasks\FigSec003`

### 3.1.3 Recommended layer sequence

| | | | |
|---|---|---|---|
| Layer 1 | Gate layer | 8 nm Cr | mask 38 at 0 deg. |
| Layer 2 | Insulator | 300 nm PaN + 80 nm PS | no mask |
| Layer 3 | Semicond | 40 nm molec. semic. | no mask |
| Layer 4 | Inject. cont. | 3 nm $MoO_x$ | mask 38 at 90 deg. |
| | Inject. cont. | 10 nm Au | mask 38 at 90 deg. |
| Layer 5 | Cont. pads | 3 nm Cr | mask 49 |
| | Cont. pads | 150 nm Ag | mask 49 |
| | Cont. pads | 20 nm Au | mask 49 |

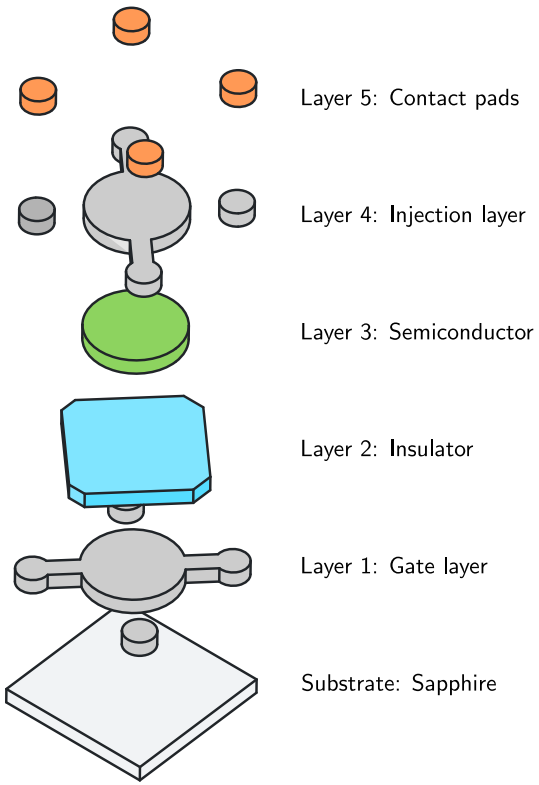



Figure 3.1: Exploded view of a standard varactor device.

## 3.2 Device layout A

### 3.2.1 Applications

- gated van der Pauw measurements
- THz measurements

### 3.2.2 File locations

**CAD file name:** `003002004.FCStd`

**Directory:**
`Q:\THz-Office\StudentOrganize\ScienceTasks\ ...`
`...\Figures\FigSec003\FreeCads\zz003002`

**Scalable vector graphics:** `yySec003002.svg`

**Directory:**
`\..\StudentOrganize\ScienceTasks\FigSec003`

### 3.2.3 Recommended layer sequence

| | | | |
|---|---|---|---|
| Layer 1 | Gate cont. | 8 nm Cr | mask 71 |
| Layer 2 | Insulator | 300 nm PaN + 80 nm PS | no mask |
| Layer 3 | Inject. cont. | 3 nm Cr | mask 72 |
| | Inject. cont. | 20 nm Au | mask 72 |
| Layer 4 | Cont. pads | 3 nm Cr | mask 70 |
| | Cont. pads | 150 nm Ag | mask 70 |
| | Cont. pads | 20 nm Au | mask 70 |
| Layer 5 | Semicond | 40 nm molec. semic. | mask 41 |

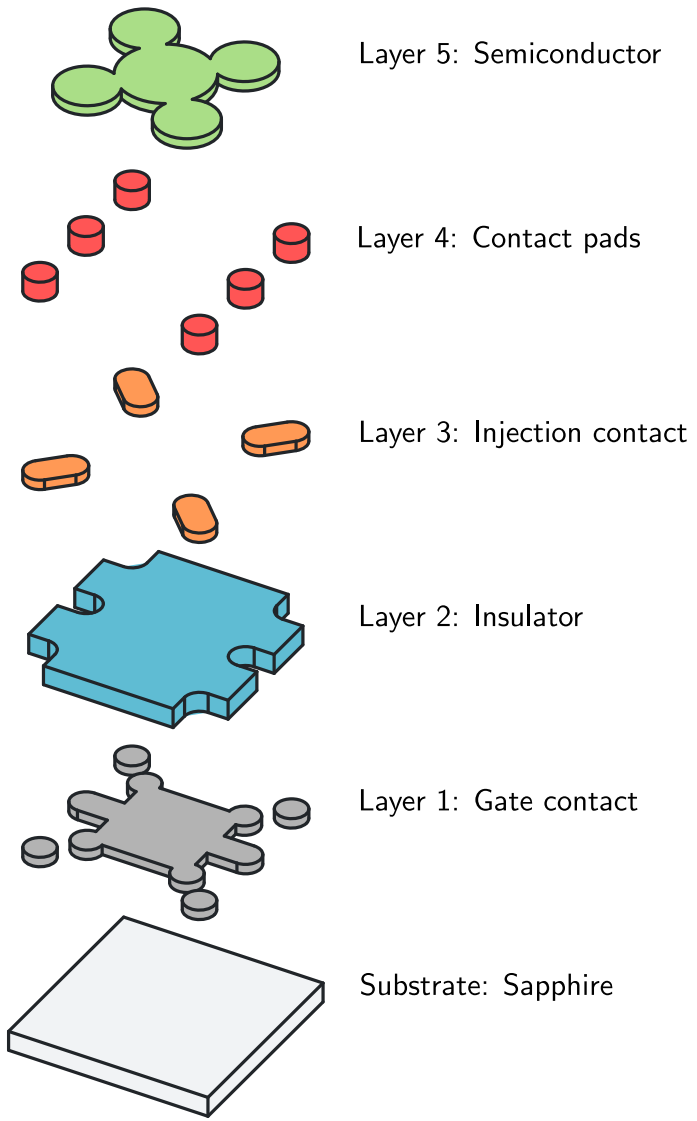



Figure 3.2: Exploded view of a device for gated van der Pauw measurements. Mind: The semiconductor is on top of the injection contatcs.

## 3.3 Device layout B

### 3.3.1 Applications

- gated van der Pauw measurements
- THz measurements

### 3.3.2 File locations

**CAD File name:** `003003001.FCStd`

**Directory:**
`Q:\THz-Office\StudentOrganize\ScienceTasks\ ...`
`...\Figures\FigSec003\FreeCads\zz003003\`

**Scalable vector graphics:** `yySec003003.svg`

**Directory:**
`\..\StudentOrganize\ScienceTasks\FigSec003`

### 3.3.3 Recommended layer sequence

| | | | |
|---|---|---|---|
| Layer 1 | Gate cont. | 8 nm Cr | mask 71 |
| Layer 2 | Insulator | 300 nm PaN + 80 nm PS | no mask |
| Layer 3 | Semicond | 40 nm molec. semic. | mask 41 |
| Layer 4 | Inject. cont. | 3 nm $MoO_x$ | mask 72 |
| | Inject. cont. | 20 nm Au | mask 72 |
| Layer 5 | Cont. pads | 3 nm Cr | mask 70 |
| | Cont. pads | 150 nm Ag | mask 70 |
| | Cont. pads | 20 nm Au | mask 70 |

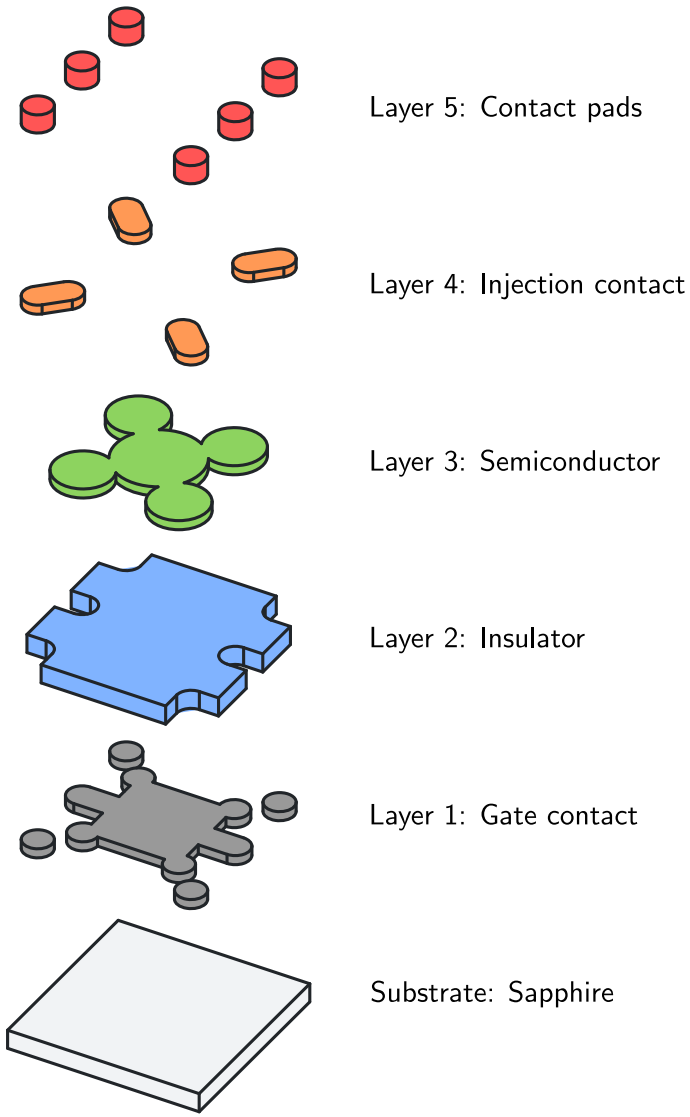



Figure 3.3: Exploded view of a device layout B. Mind: The injection contatcs are on top of the semiconductor.

## 3.4 Dual-line-package in THz Chamber

- The schematic shows the electrical connection of the varactor and side-injection devices to a dual-in-line package (DIP) inside the THz chamber.
- Two wiring configurations are illustrated: one for six-pin devices and one for four-pin devices.

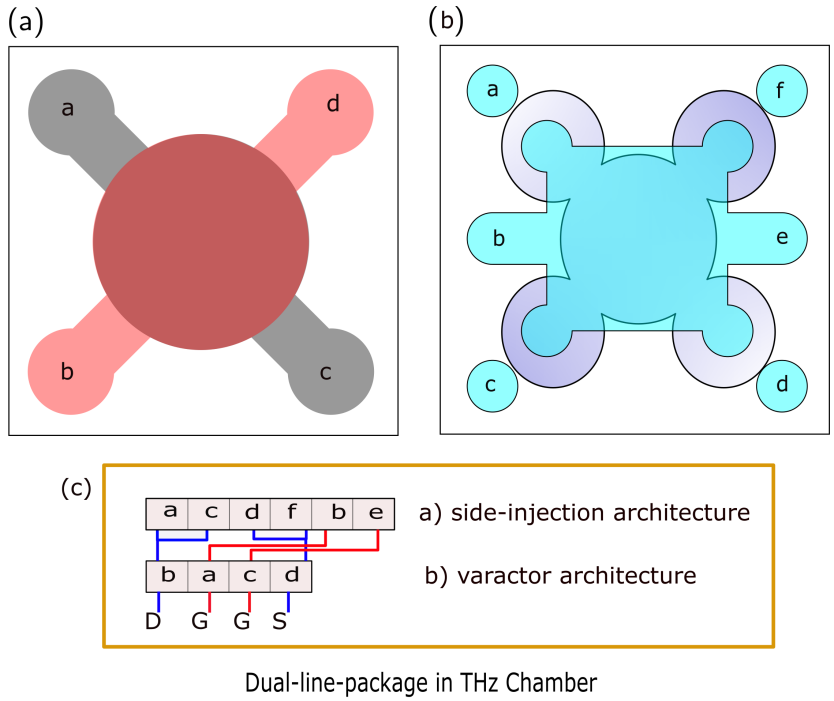



Figure 3.4: Wiring of the varactor and side-injection devices to the dual-in-line package (DIP) in the THz chamber for biasing and signal access.

# 4 Cleaning

**Date of issue:** 13.08.2025

**General:**

- Yes, lab cleaning belongs to the scientific tasks. Those who are using the lab, are supposed to clean it, too.
- A general clean-up should be done every three months.

## 4.1 Procedure for general lab cleaning

1. Store all items that can be stored in the cabinets.
2. Cleaning the tops of the hoods, cabinets, and tubes:
   - will be added later
3. Cleaning the top-compartments of the hoods:
   - will be added later
4. Cleaning the floor:
   - Vacuum the floor of the lab.
   - Remove dust with the N2-guns.
   - Wet clean the floor.
5. Cleaning under the hoods:
   1. Remove dust with the N2-guns.
   2. Uninstall the drapes (windows). Clean them on both sides with window cleaner.
   3. Wet-clean the tables.
   4. Wet cleaning of the optical table: Make sure that you don't touch mirrors or optics.

### 4.1.1 Last general clean-ups

- next date
- September 2025
- December 2024
- July 2024

## 4.2 Procedure for cleaning the PVD

The PVD should be cleaned on demand.

### 4.2.1 Last cleaning of the PVD

- next date
- September 2025
- December 2024
- February 2024

## 4.3 Procedure for cleaning the CVD

The CVD should be cleaned on demand.

### 4.3.1 Last cleaning of the CVD

- next date
- September 2025
- June 2025?

# 5 Thickness of MoOx

**Date of issue:** 20.08.2025

**Last changes**

- 20.08.2025, RK: setup uf document

## 5.1 General remarks

### 5.1.1 Background

- $MoO_x$ is a suitable material for hole injection into molecular semiconductors and polymers. The conduction band edge of $MoO_x$ is close to the valence band edge of many organic semiconductors.
- We have to find literature on $MoO_x$ used for injection. Please cite references here.

### 5.1.2 Understanding of the injection layer

- For the injection layer, we usually deposit 2 nm $MoO_x$ plus 12 nm Cr.
- However, both layers should not be understood as beeing homogeneous as shown in Fig. 5.1 (a).
- The $MoO_x$ may form clusters with areas in between, where the $MoO_x$ doesn't cover the semiconductor surface Fig. 5.1 (b). In these gaps the Cr-layer can stick to the organic semiconductor

### 5.1.3 Hypothesis

If the $MoO_x$ covers the entire surface of the semiconductor Fig. 5.1 (c), the Cr-layer won't stick to the semiconductor. Because $MoO_x$ is very brittle, the Cr-Layer can peel off.

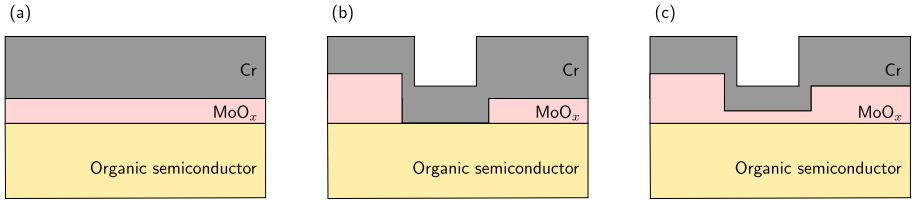



Figure 5.1: Models for the Topology of the injection layer. For graphical purposes, the roughness of the semiconductor surface is not considered. (a) Unrealistic model, which assumes flat surfaces of the $MoO_x$ and of the Cr-layers. (b) Clustering of $MoO_x$, but the surface of the semiconductor is not fully covered. (c) The $MoO_x$layer covers the entire surface of the semiconductor.

### 5.1.4 Central question

- How thick should the $MoO_x$ layer be?
- If the $MoO_x$ layer is too thin, injection might be poor?
- If the $MoO_x$ layer is too thick, the $MoO_x$ plus the Cr-Layer may peel off.

## 5.2 Experiment and results

### 5.2.1 Experiment

- Use a Teonex foil, the organic side will be used for simulating an organic semiconductor.
- Deposit the $MoO_x$ layer, but along the surface a gradient in thickness is desired:
  - Use program `MetalPCD004021`.
  - Start process
  - When a decent sublimation rate is reached, start deposition.
  - Every thickness increase by 0.5 nm close the shutter by 10 degree.
- Then deposit 12 nm Cr, homogeneously over the $MoO_x$.
- After deposition look at which thickness of the $MoO_x$ the Cr-layer peels off:
  - take a photo
  - include in figure here

### 5.2.2 Results

## 5.3 Conclusion

# 6 Installing LaTeX & Co.

**Date of issue:** 18.09.2025

**Last changes**

- 18.09.2025, RK: setup uf document

## 6.1 Fundamentals

All installtion software and programs are provided by the Folder `AllForLaTex`. You may find the folder under `Q:\THz-Office`

All descriptions are exclusively meant for members of the THz-Group.

### 6.1.1 Remarks on LaTeX

- LaTeXis supplied by the MikTeX distribution.
- MikTex should be installed locally, which means every user has to install it under his/her user account. Then the MikTeX sofware will be automatically installed, for instance, under
  `C:\User\Kersting.lokal\AppData\Local\Programs\MikTeX`.
- Installation of the MikTeX package as an administrator is not recommended. Such installations frequently collapse.

### 6.1.2 Remarks on Ghostview

- Ghostview is perfect if you want to have a quick view onto your document. Use it during texting.
- For the very final output you may want to generate a pdf-file, but this takes much longer than working with postscript.

### 6.1.3 Remarks on the LaTeXshell by RK

- This is a very small GUI, which provides buttons for:
  - selecting a LaTeX-file,
  - displying this file in an editor,
  - starting LaTeX(not pdf-Latex or other fancy stuff), which will generate a dvi-file,
  - starting bibtex,
  - to make a ps-file from the generated dvi-file,
  - to make a pdf-file from the generated dvi-file,
  - opening Ghostview to have a look onto your document,
  - and some other less important functions.
- Within the THz-Group the LaTeXfiles should be compatible and exchangeable. Thus, it is suggested to use this shell and pure LaTeX.

### 6.1.4 Remarks on the Boxer editor

- Many other editors may be suited. When using the suggested LaTeX shell the only requirement is that these editors can be started by a DOS command, such as `MyEditor.exe MyFile.tex`.
- Use of the Boxer editor is recommended. Licences can be obtained from `boxersoftware.com`.
- The files in All4LaTeX are only meant as examples and should not be installed.

### 6.1.5 Remarks on TexWorks

- So far, this programs doesn't include eps-files properly when running in this shell pdflatex.
- For coherence within the group, the policy is to use eps-files for LaTeX.

## 6.2 Installation

### 6.2.1 Preparations

- Copy the folder `Q:\THz-Office\All4LaTeX` to your desktop.
- Sorry, but once the folder is here, it has to stay here als long as you use the libraries under `All4LaTeX\LocalTex`.

### 6.2.2 Installing MikTeX

- Make sure that you are logged in as the user for whom you want to install MikTeX.
- Goto folder `All4LaTeX\MikTeX` and run `basic-miktex-24.1-x64.exe`.
- Install as a user – not as an administrator!
- This installs the MikTex package, for instance, under `C:\User\Kersting.lokal\AppData\Local\Programs\MikTeX`.

### 6.2.3 Installation of Ghostview

- Deinstall all GhostView and GhostScript software from your computer.
- The new installation files can be found on your desktop in `All4LaTeX`.
- The sequence of installtions is important:
  - First, install `gs927w64.exe`.
  - Second, install`gsv50w64.exe`.
- Run Ghostview and set the following parameters (which should be adapted to your paths)
- Konfiguration: 9.53
- Fortgeschrittene Konfigurierung:
  - Ghostscript-Dll:
    `C:\Program Files\gs\gs9.27\bin\gsdll64.dll`
  - Ghostscript-Suchpfad:
    `C:\Program Files\gs\gs9.27\bin;`
    `C:\Program Files\gs\gs9.27\lib;`

C:\Program Files\gs\gs9.27\fonts

- Ghostscript-Optionen:
  -dNOPLATFONTS -sFONTPATH="c:\psfonts"

Mind: Updates of GhostView and/or GhostScript should be avoided.

### 6.2.4 Setting up the LaTeX shell

- It is supposed that the folder All4LaTeX is located on you desktop.
- Find ShellLaTeX06_09.exe.
- Make a shortcut and copy this shortcut to your desktop.
- It is important that the original ShellLaTeX06_09.exe stays in the folder All4LaTeX\LaTeX shell
- In All4LaTeX\LaTeX shell, open the parameter file ShellParm01.dat
- You will see the paths that direct the shell to the programs.
- Check these paths and make changes if necessary.

### 6.2.5 Personalizing MikTeX

This section describes how to make MikTex know about your personal libraries, selfwritten makros etc.

- All libraries and makros you want to use should be on your desctop in All4LaTeX.
- Run the MikTeX Console (Windows symbol, programs, and find MikTex).
  - Avoid updates. Usually, this doesn't make things better, but may crash your system.
  - On the left go to Einstellungen.
  - Select the tab Verzeichnisse
  - Add here a new entry, for instance, All4LaTex\LocalTex, which should be located on your desktop.
  - Goto to top-line tab Aufgaben and select Dateinamendatenbak synchronisieren
  - Close the MikTex console.

# 7 Side-injection device for THz and gvdP

**Date of issue/modification:** 22.09.2025

## 7.1 General

### 7.1.1 Background

1. If side injection were possible:
   - device facrication would become much more simple
   - there is no need for cooling during deposition of the top contacts
   - see discussion in sec. 2
2. If THz and gvdP measurements can be performed
   - macroscopic and microscopic transport can be compared
   - degradation of THz devices can be probed by gvdP

### 7.1.2 Questions

1. Is side injection possible?
2. Is it possible to build devices that serve for THz measurements as well as vor gvdP measurements?
3. What is an optimized fabrication path?

## 7.2 Layout A

### 7.2.1 Properties

- see Fig. 7.1 and Tab. 7.1
- This layout should allow for THz and for gvdP measurements.
- It uses side-injection of the charge carrieres.
- The semiconductor layer is deposited last.
  - Thus different semiconductors can be deposited to the same electronic structure.
  - The expensive semiconductor material is less wasted, because its the last material to be deposited.

Table 7.1: Process parameters for layout A.

| Layer | Purpose | Mask # | Material | Process |
|---|---|---|---|---|
| 1 | Substrate | – | sapphire | – |
| 2 | Gate metal | 71 | 7 nm Cr | PVD |
| 3 | Insulator | – | 250 nm PaN + 70 nm PS | CVD and spin coating |
| 4 | Injection layer | 72 | 2 nm Cr + 10 nm Au | PVD |
| 5 | Contact pads | 70 | 2 nm Cr + 150 nm Ag + 40 nm Au | PVD |
| 6 | Semiconductor | 41 | 35 nm molec. semicond | PVD |

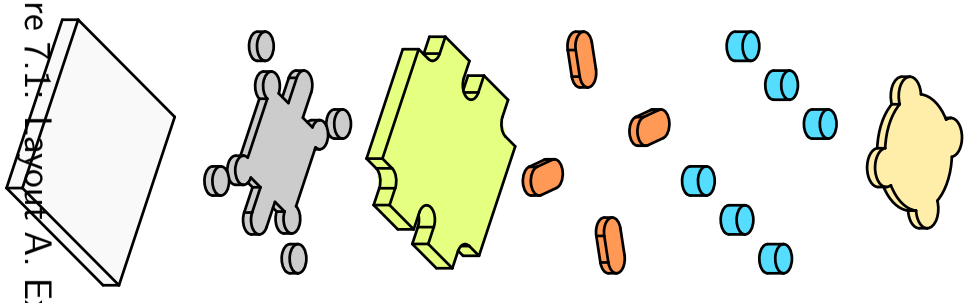



Figure 7.1: Layout A. Exploded view of the individual layers.

## 7.3 Simulation

**Situation:**

- injection contact bottom, semiconductor top
- see Fig. 7.3

**Question and answers:** Which properties has the charge distribution?

- charge injection works
- in the vicinity of the contact the charge density is reduced, because of band offsets
- the screening length is about 1 nm

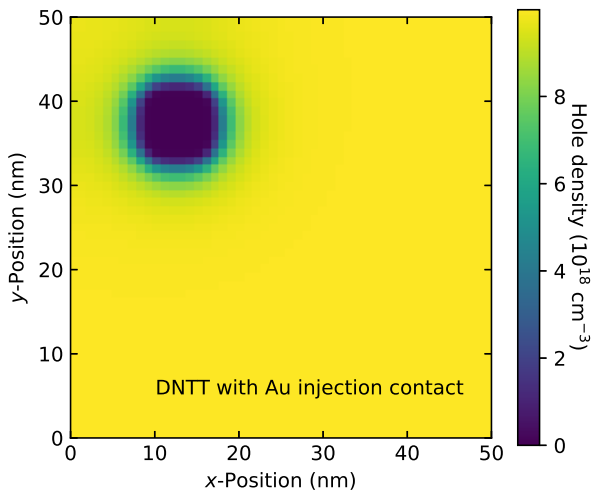



Figure 7.2: Charge distribution around an injection contact made of gold.

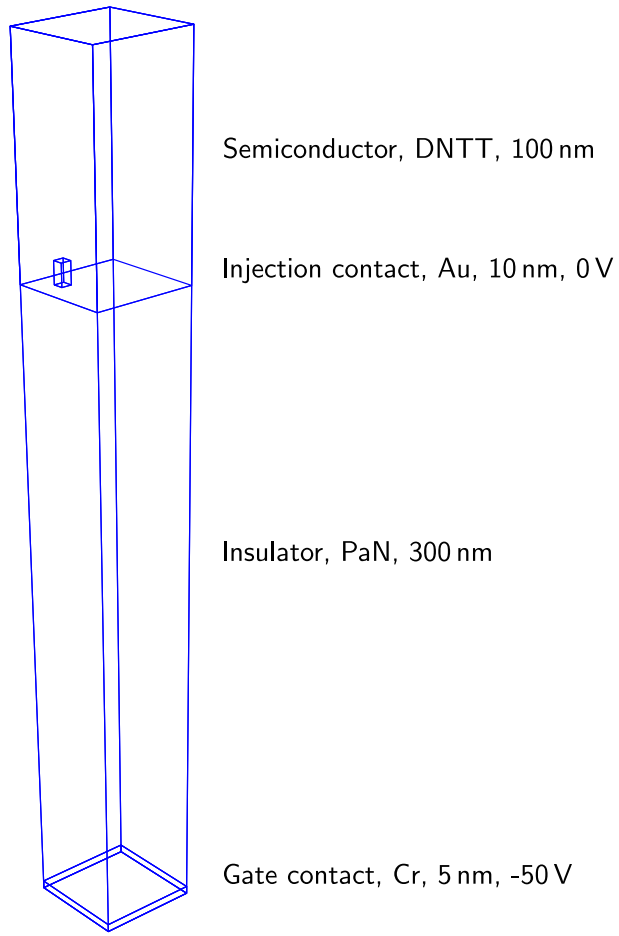



Figure 7.3: Simulated structure.

# 8 Capacitance-voltage technique

**Date of issue:** 02.10.2025, RK

## 8.1 Background

Many techniques provide the sheet conductance $\sigma_{2D}$ with $[\sigma_{2D}] = 1\Omega$. These are

- THz electromudulation experiments,
- van der Pauw-measurements, and
- measurements on field-effect transistors.

The sheet conductance $\sigma_{2D}$, however, is given by the carriers' sheet density $n_{2D}$ and their mobility $\mu$:

$$\sigma_{2D} = e\, n_{2D} \mu \tag{1}$$

**Mind:**

- The above equation assumes a homogeneous carrier sheet.
- The carrier sheet density $n_{2D}$ is a property of an individual measurement.
- The mobility $\mu$ is a material property.

**Goal:** To deduce $n_{2D}$ with another technique, operated under similar conditions.

**Suitable methods:** The carriers' sheet density can be determined by methods, such as:

- Hall measurements,
- CV-measurements,
- recording the charging current in time domain (similar to DLTS)
- ...

## 8.2 Capacitance-voltage measurements

### 8.2.1 Definition

So called CV measurements provide the dependence of the device's capacitance $C$ on the applied voltage $V$, i.e. $C(V)$.

### 8.2.2 Importance for our work

- Usually, an applied voltage causes a sheet carrier density. If both $C$ and $V$ are known then:

$$n_{2D} = e\,\tilde{C}\tilde{V} = e\,\frac{C}{A}\tilde{V} = \tag{2}$$

  where $A$ is the capacitors area.
- In some devices the applied voltage introduces a geometrical distribution of charge carriers, for instance in field-effect devices with side injection contacts. Usually, the area $A$ of this distribution is unknown, but becomes accessible by measuring $C$. Again, the ultimate goal is to obtain $n_{2D}$.

### 8.2.3 Impedance measurements

The measurement principle is shown in Fig. 8.1:

- An AC source provides $V_{AC}$.
- The capacitance is charged via load resistor $R_L$ and probe resistor $R_P$.
- The voltage over $R_P$ is measured ($V_{LIA}$).

The total impedance of the circuit is:

$$X_{tot} = R_L + R_P + \frac{-i}{\omega C} \tag{3}$$

The LIA probes the voltage over the voltage divider:

$$V_{LIA} = \frac{R_P}{R_L + R_P + X_C} \cdot V_{AC} \tag{4}$$

Solve for capacitance:

$$\begin{aligned}
\eta &= \frac{V_{LIA}}{V_{AC}} = \frac{R_P}{R_L + R_P + X_C} && (5)\\
R_L + R_P + X_C &= \frac{R_P}{\eta} && (6)\\
X_C &= \frac{R_P}{\eta} - R_L - R_P && (7)\\
\frac{-i}{\omega C} &= -\left(R_L + R_P\left(1 - \frac{1}{\eta}\right)\right) && (8)\\
C &= \frac{i}{\omega} \cdot \frac{1}{R_L + R_P\left(1 - \frac{1}{\eta}\right)} && (9)
\end{aligned}$$

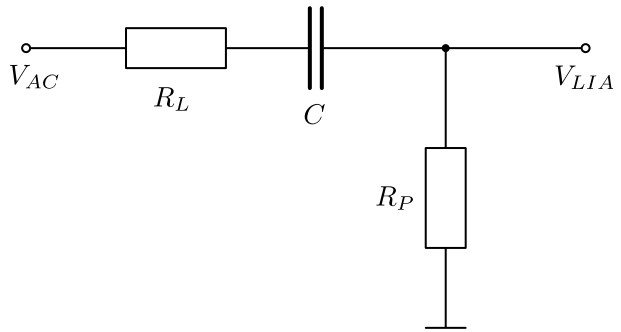



Figure 8.1: Layout for measuring the capacitance $C$.

### 8.2.4 Measurement circuit

The final measurement circuit is shown in Fig. **??**. It comprises:

- The superposition of $V_{AC}$ and $V_{DC}$ via $R_3$ – $R_5$.
- Buffer amplifiers at the inputs of $V_{AC}$ and $V_{DC}$.
- A summing amplifier for $V_{AC}$ and $V_{DC}$.
- An inverter.

### 8.2.5 Required devices

1. a 15-V pover supply for the measurement circuit
2. a SR830 lock-in
   - for supplying $V_{AC}$
   - for detecting the signals
3. a NI6001, which supplies $V_{DC}$
4. an amplifier, which may be inserted just before $R_L$

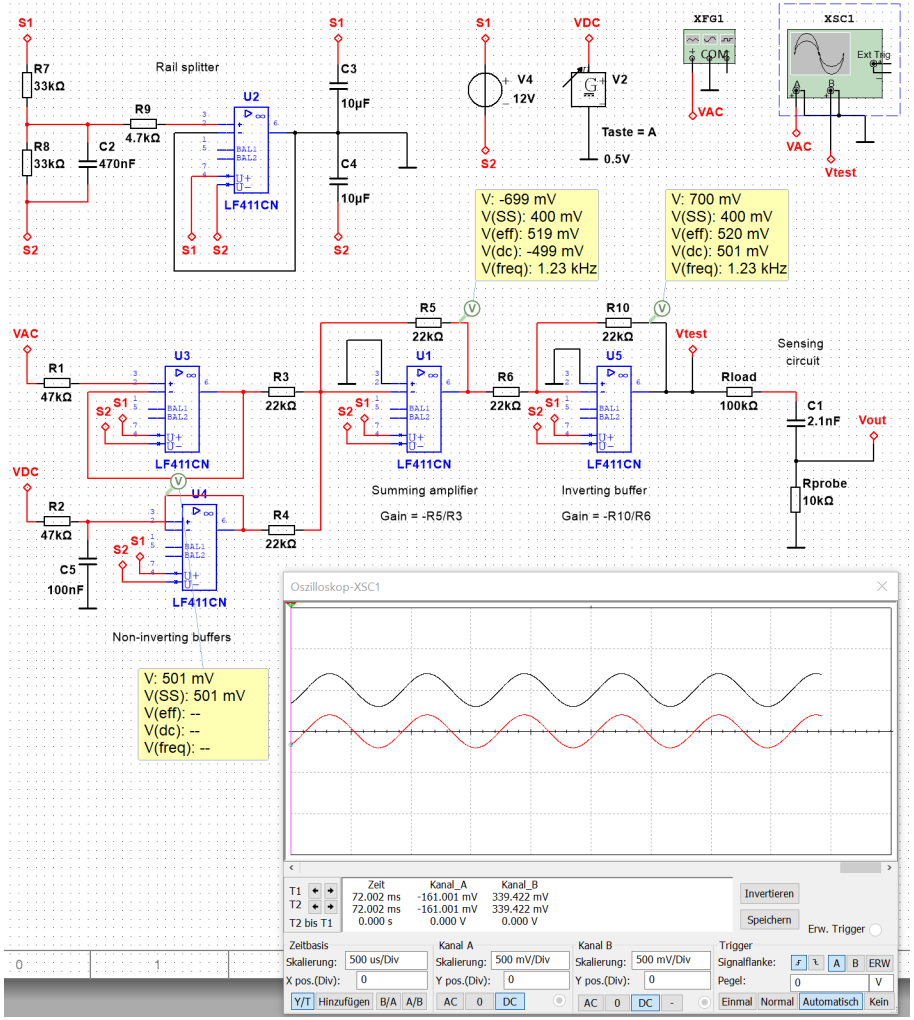



Figure 8.2: Measurement circuit.

## 8.2.6 Software

- as of 02.10.25: program `CapLia001034.vbp`
- in `Q:\THz-Lab\Software Source Code\ ...`

`...\Software Measurement\ElectricalCharact\CapLia_202`

## 8.2.7 Batch file for scan

| | | |
|---|---|---|
| 1 | RK | Initials: |
| 2 | 20221028 W06-01 | Sample name: |
| 3 | C8-BTBT-C8 | Comment 1: |
| 4 | xxxx | Comment 2: |
| 5 | xxxx | Comment 3: |
| 6 | xxxx | Comment 4: |
| 7 | -666 | void: |
| 8 | -666 | void: |
| 9 | -666 | void: |
| 10 | -666 | void: |
| 11 | 0 | Simulation (0: FALSE, 1: TRUE): |
| 12 | 334.12 | LIA output freq. (Hz): |
| 13 | 0.01 | LIA output ampl. (V): |
| 14 | 0.30 | LIA time constant (s): |
| 15 | 0.02 | LIA sensitivity (V): |
| 16 | -666 | void: |
| 17 | -666 | void: |
| 18 | -666 | void: |
| 19 | -666 | void: |
| 20 | -666 | void: |
| 21 | 99 | Repetitions to do (1): |
| 22 | 1 | Bidirectional (0: FALSE, 1: TRUE): |
| 23 | 51 | DC voltage bins (1): |
| 24 | 2.0 | DC voltage min. (V): |
| 25 | -2.0 | DC voltage max. (V): |
| 26 | 2 | Reps. before auto save (1): |
| 27 | 3 | Meas. before plot (1): |
| 28 | 300 | Fundamental wait (ms): |
| 29 | -666 | void: |
| 30 | -666 | void: |
| 31 | -666 | void: |
| 32 | 1.6e5 | Load resistor (Ohms): |
| 33 | 1e4 | Probe resistor (Ohms): |
| 34 | 1 | Ampl. fact. (1 or 20 or -20): |
| 35 | 25e-6 | Area of capacitor in unit m2: |
| 35 | -666 | void: |
| 37 | -666 | void: |
| 38 | -666 | void: |
| 39 | -666 | void: |

## 8.3 Examples

### 8.3.1 Example on mica capacitor

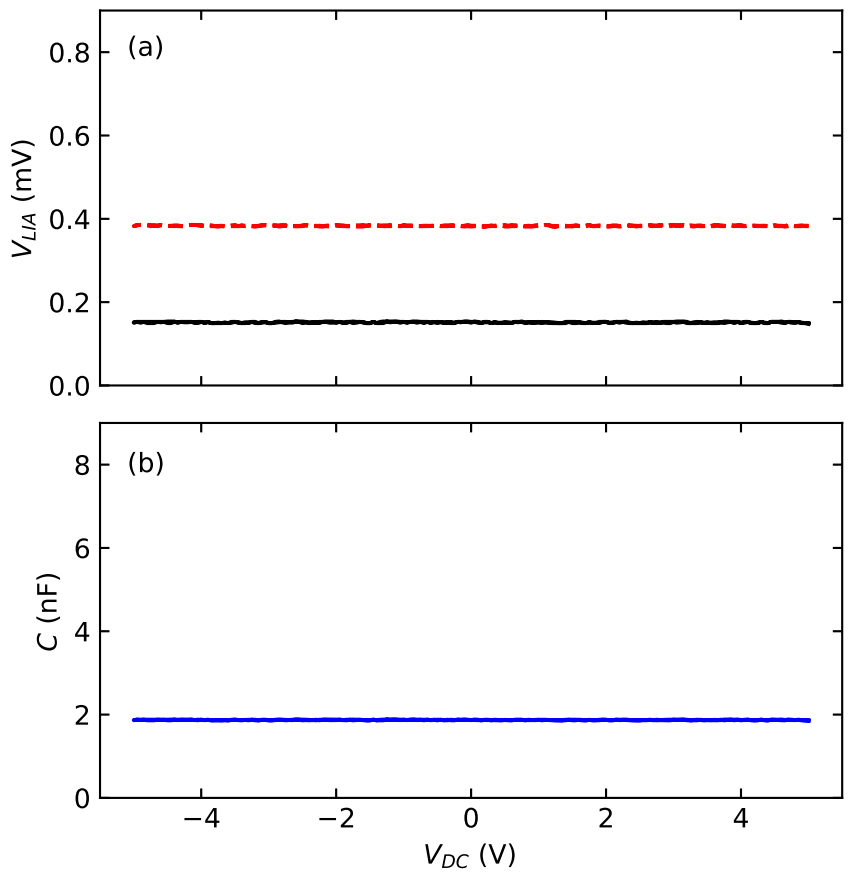



Figure 8.3: Measurement recorded on a mica capacitor.

### 8.3.2 Example on 2.1 nF capacitor

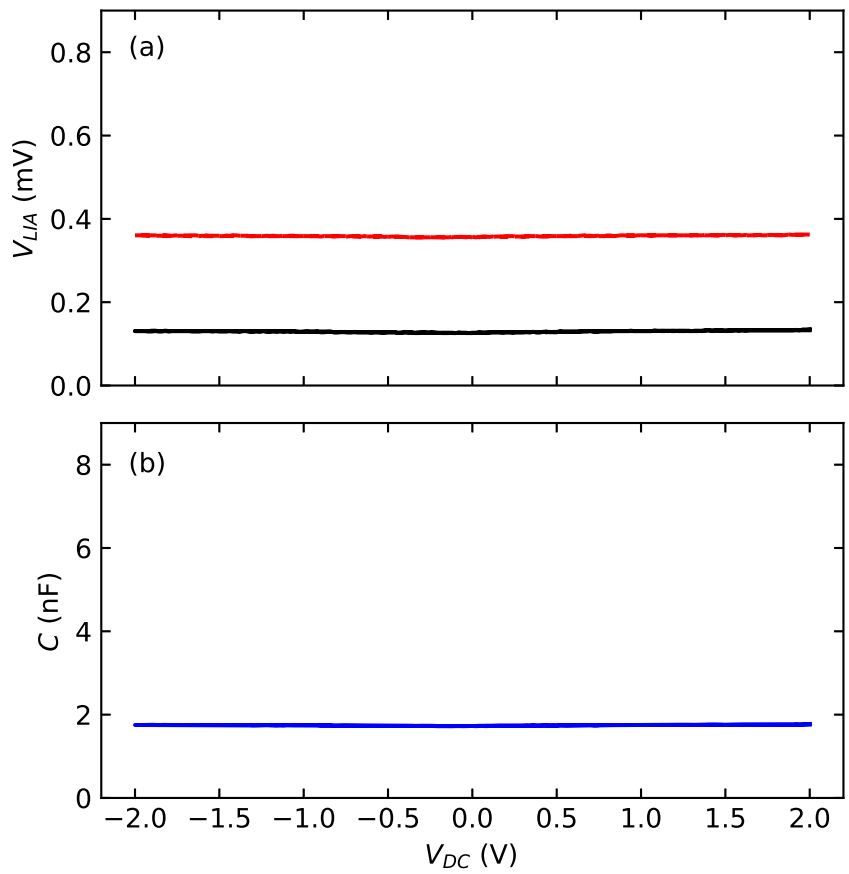



Figure 8.4: Measurement recorded on a conventional capacitor.

### 8.3.3 Example on BAT128

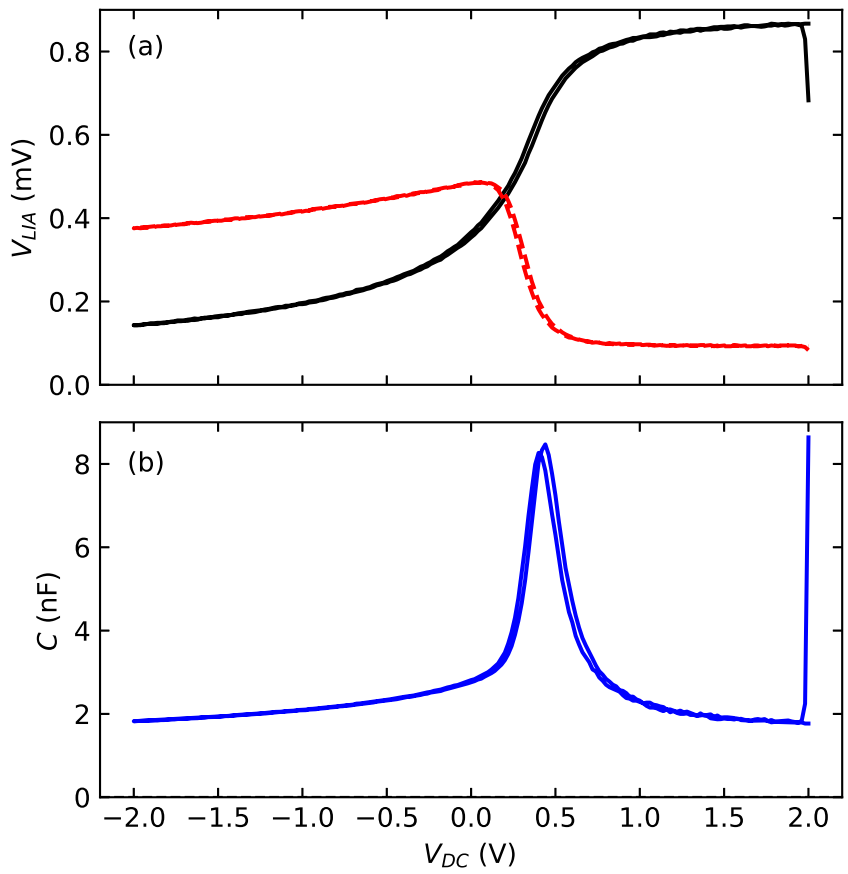



Figure 8.5: Measurement recorded on BAT128.

## 8.4 Influence of BNC Cable Length on Capacitance Measurement

- more than 1m cable → 1 nF
- 1 m cable → 0.5 nF
- 0.5 m cable → 70 pF
- 0.25 m cable → 50 pF
- The capacitance of the holder → 40 pF

## 8.5 Calculation of the Capacitor area

| Parameter | Area ($m^2$) | Capacitance (nF) |
|---|---|---|
| $C_{\text{big}}$ | $A_{\text{big}} = 32 \times 10^{-6}$ | $C_{\text{big}} = 2.6$ |
| $C_{\text{small}}$ | $A_{\text{small}} = 1.5 \times 10^{-6}$ | $C_{\text{small}} = 0.2$ |

$C_{\text{big}}$ =Intersection between Gate and Semiconductor
$C_{\text{small}}$=Intersection between Gate and Injection

$$Area_{Sc}=5^2+4*p(1.5/2)^2$$

$$Area_{Sc}=32mm^2$$

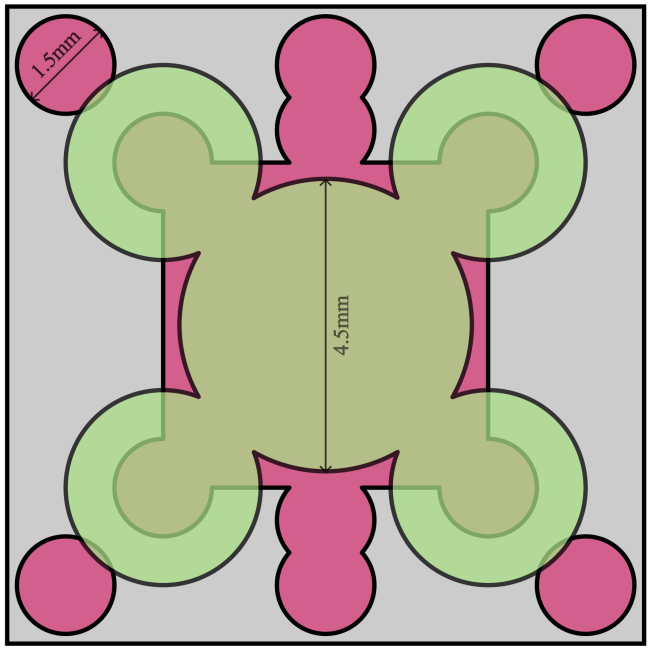



Figure 8.6: Calculation of the Capacitor area.

# 9 PVD Deposition Parameters

**Date of issue:** 15.10.2025

- Metals: Ag, Cr, MoOx, Au, etc.
- Organic semiconductors: C8-BTBT, etc.
- Organic insulators: TripC12, etc.

The main goal is to provide a standardized refere
parameters for different materials used in device

## 9.1 Metal Deposition Parameters

### 9.1.1 Silver (Ag)

**PVD Parameters for Silver (Ag)**

| Step | Ramp 1 (A) | Soak 1 (s) | Ramp 2 (A) | Sc |
|---|---|---|---|---|
| 1 | 15 | 60 | 25 | |

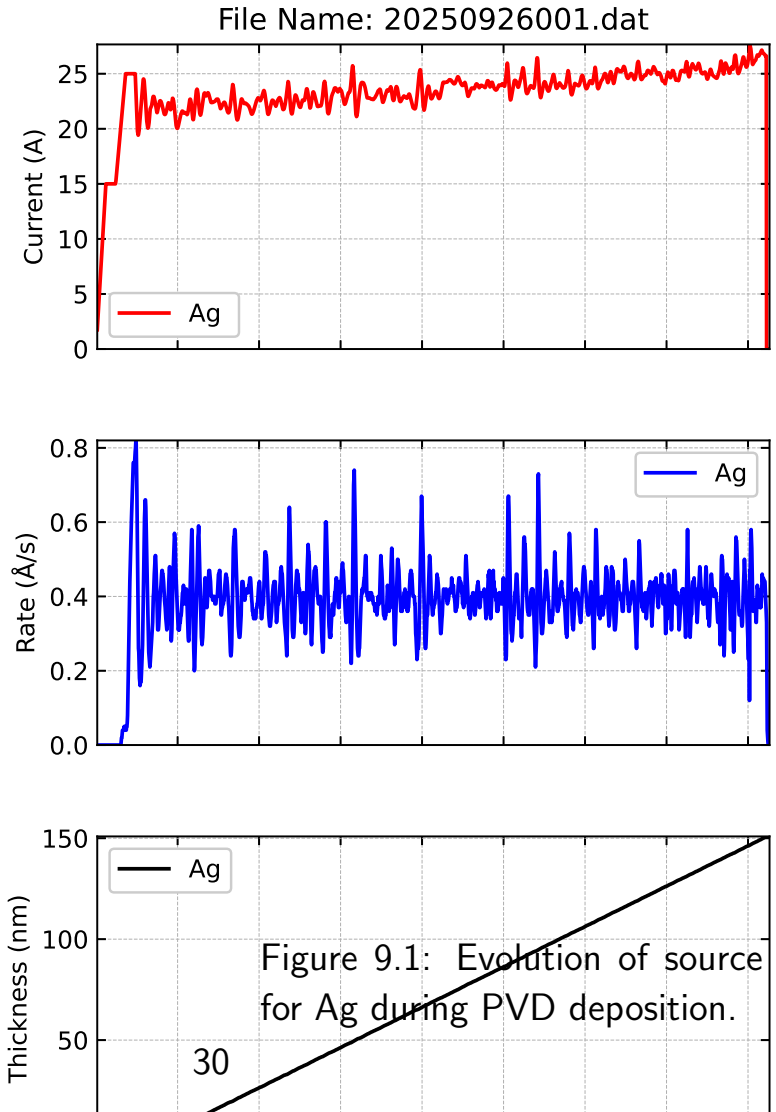



Figure 9.1: Evolution of source current and film thickness for Ag during PVD deposition.

### 9.1.2 Gold (Au)

Table 9.1: PVD Parameters for Gold (Au)

| Step | Ramp 1 (A) | Soak 1 (s) | Ramp 2 (A) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 20 | 60 | 30 | 60 |

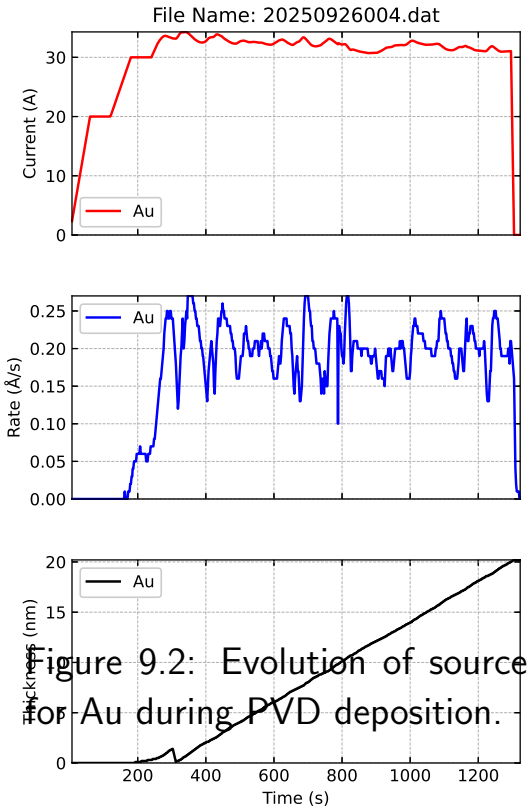



Figure 9.2: Evolution of source current and film thickness for Au during PVD deposition.

### 9.1.3 Chromium (Cr)

Table 9.2: PVD Parameters for Chromium (Cr)

| Step | Ramp 1 (A) | Soak 1 (s) | Ramp 2 (A) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 15 | 60 | 23 | 60 |

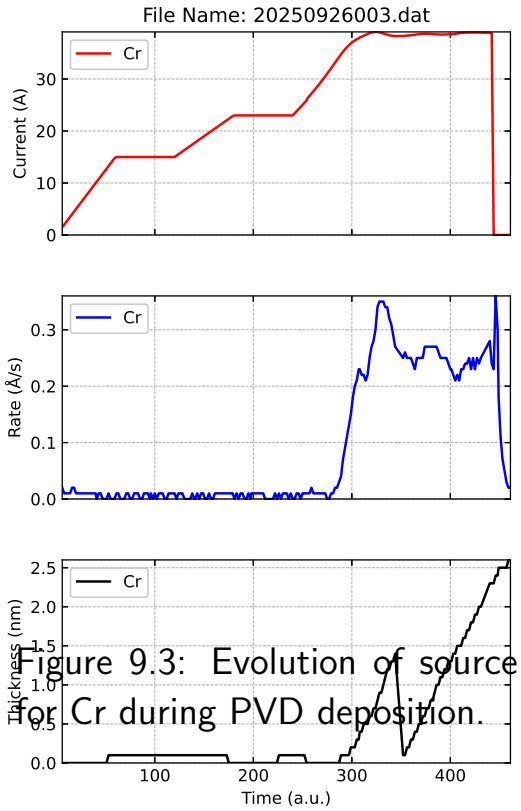



Figure 9.3: Evolution of source current and film thickness for Cr during PVD deposition.

### 9.1.4 Molybdenum Oxide (MoOx)

Table 9.3: PVD Parameters for Molybdenum Oxide (MoOx)

| Step | Ramp 1 (A) | Soak 1 (s) | Ramp 2 (A) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 15 | 60 | 17 | 60 |

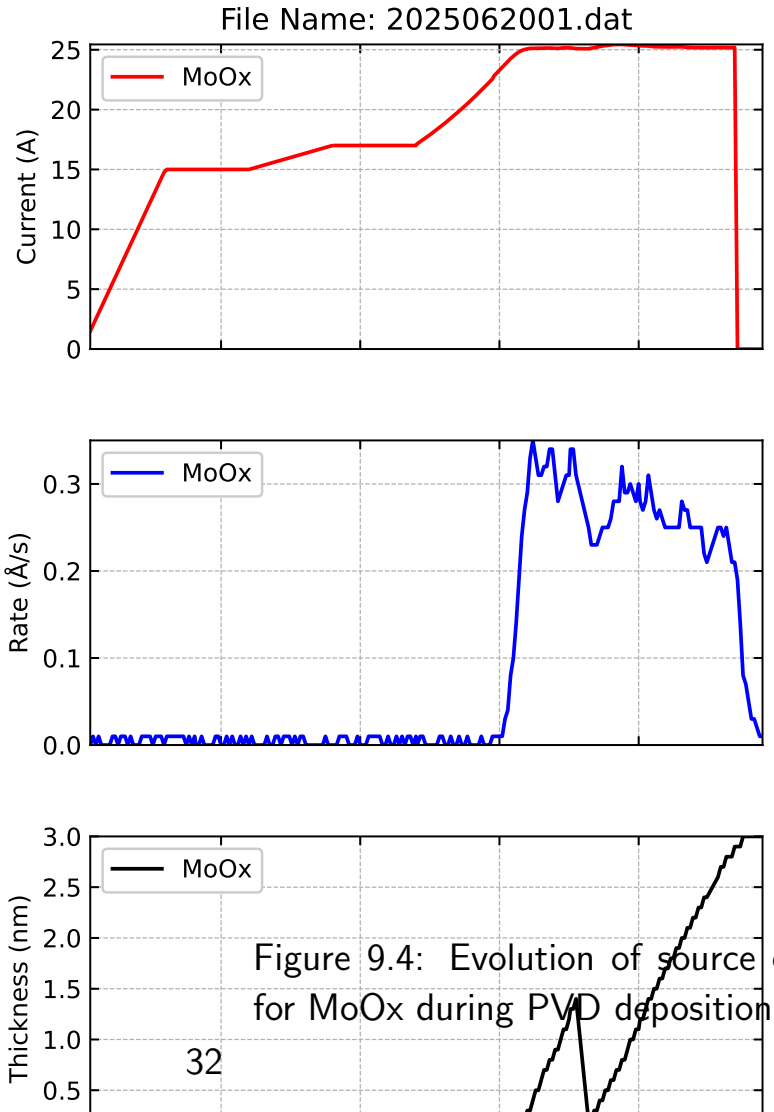



Figure 9.4: Evolution of source current and film thickness for MoOx during PVD deposition.

## 9.2 Organic Semiconductor Deposition Parameters

### 9.2.1 C8-BTBT-C8

Table 9.4: PVD Parameters for C8-BTBT-C8

| Step | Ramp 1 (°C) | Soak 1 (s) | Ramp 2 (°C) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 120 | 2000 | 150 | |

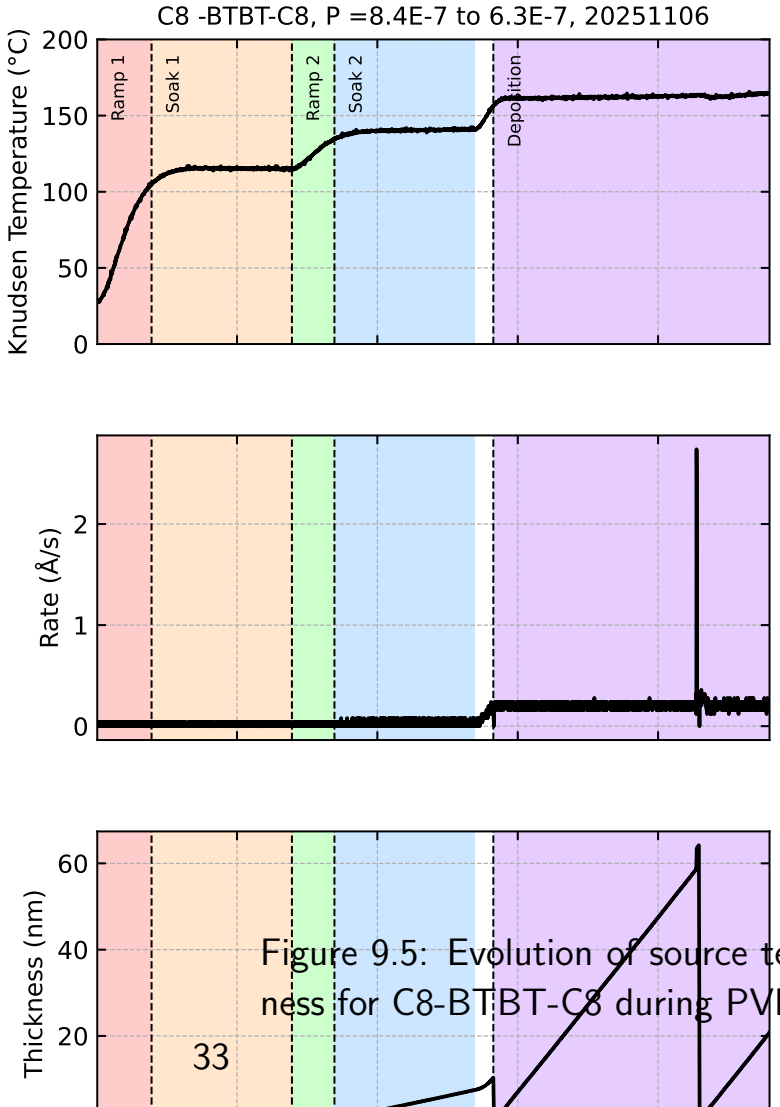



Figure 9.5: Evolution of source temperature and film thickness for C8-BTBT-C8 during PVD deposition.

### 9.2.2 Ph-BTBT-C8

Table 9.5: PVD Parameters for Ph-BTBT-C8

| Step | Ramp 1 (°C) | Soak 1 (s) | Ramp 2 (°C) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 120 | 2000 | 165 | 2000 |

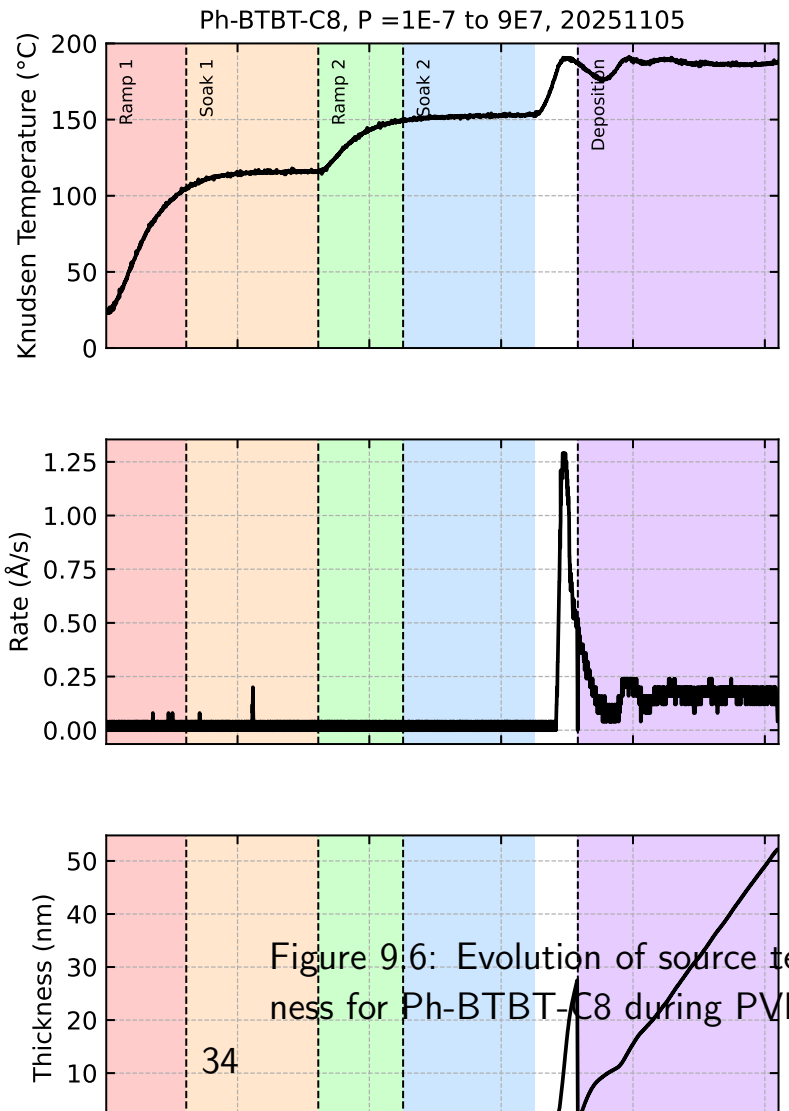



Figure 9.6: Evolution of source temperature and film thickness for Ph-BTBT-C8 during PVD deposition.

### 9.2.3 TripC12

Table 9.6: PVD Parameters for TripC12

| Step | Ramp 1 (°C) | Soak 1 (s) | Ramp 2 (°C) | Soak 2 (s) |
|---|---|---|---|---|
| 1 | 100 | 2000 | 190 | 2000 |

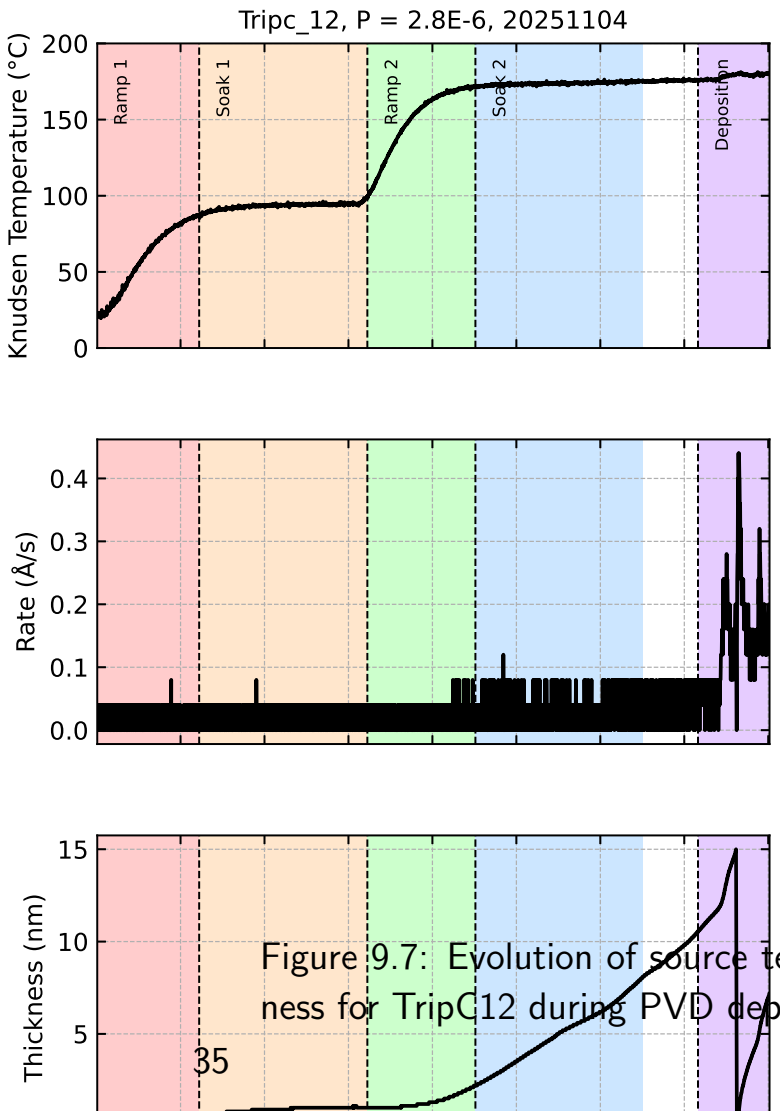



Figure 9.7: Evolution of source temperature and film thickness for TripC12 during PVD deposition.

## 9.3 Notes and Expansion

- Ramp and soak values are material-dependent and should be updated as new data becomes available.
- Metals use source current as the control parameter, organics use Knudsen cell temperature.

# 10 Electrical Characterization

**Date of issue:** 17.11.2025

## PaN Sample: 20251030_13

### IV Measurements

**Green wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/IV/Green`
- File numbers: `20251117001.dat` (-20V), `20251117002.dat` (-30V)
- Python: `20251117001.py`, `20251117002.py`

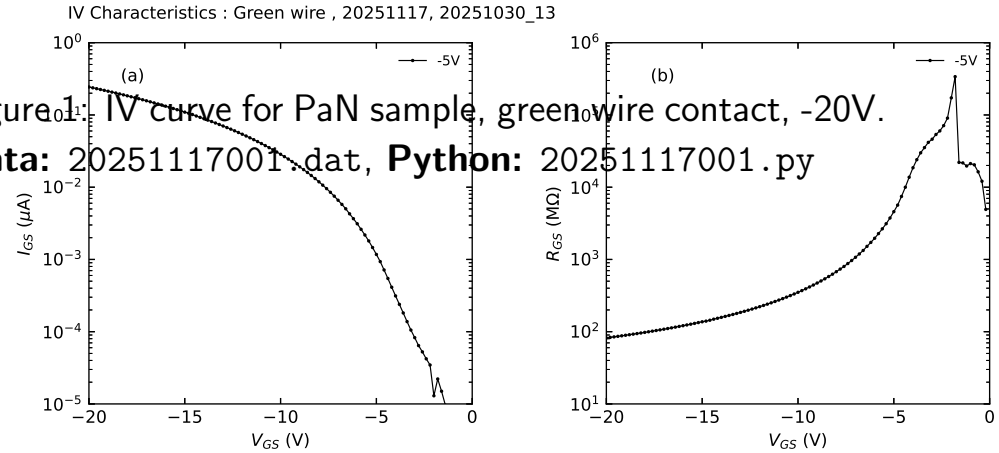



Figure 1: IV curve for PaN sample, green wire contact, -20V.
**Data:** `20251117001.dat`, **Python:** `20251117001.py`

**Red wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/IV/Red`
- File numbers: `20251117003.dat` (-20V), `20251117004.dat` (-30V)
- Python: `20251117003.py`, `20251117004.py`

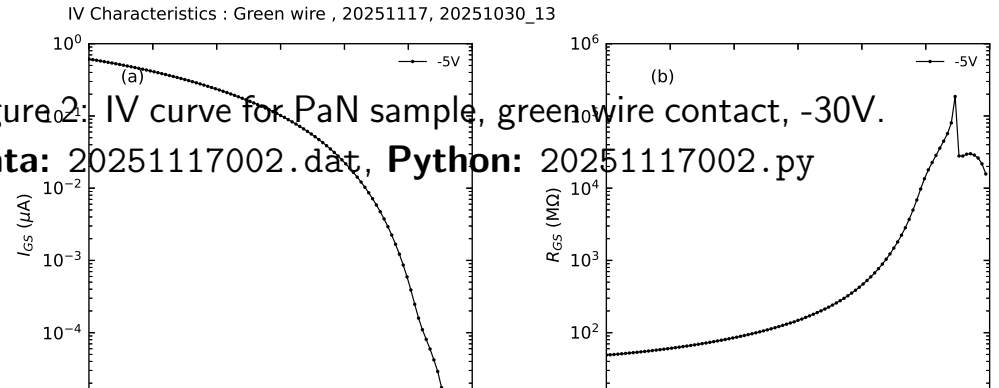



Figure 2: IV curve for PaN sample, green wire contact, -30V.
**Data:** `20251117002.dat`, **Python:** `20251117002.py`

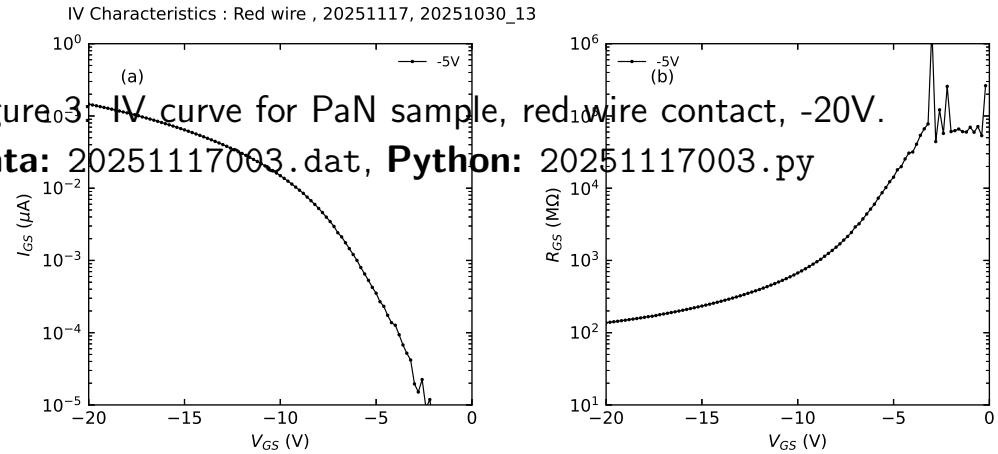



Figure 3: IV curve for PaN sample, red wire contact, -20V.
**Data:** `20251117003.dat`, **Python:** `20251117003.py`

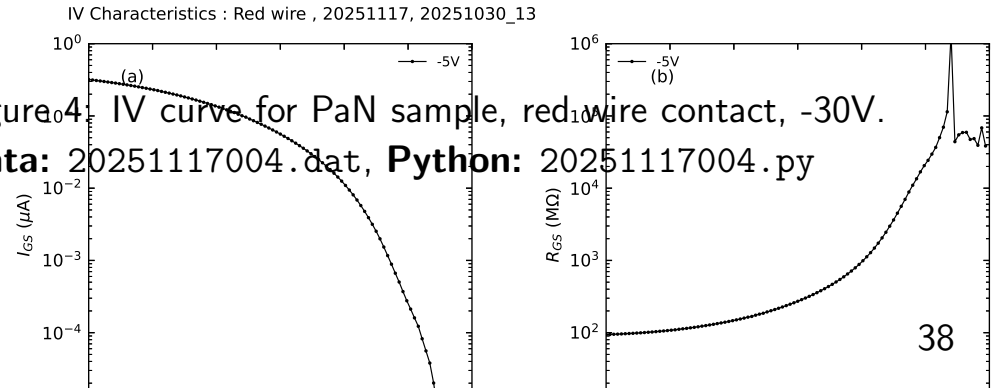



Figure 4: IV curve for PaN sample, red wire contact, -30V.
**Data:** `20251117004.dat`, **Python:** `20251117004.py`

## Capacitance Measurements (LIA)

### Green wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/CapLIA/Gre`
- File numbers: `20251117001.dat (20V to -20V)`, `20251117002.dat (30V to -30V)`
- Python: `20251117001.py`, `20251117002.py`

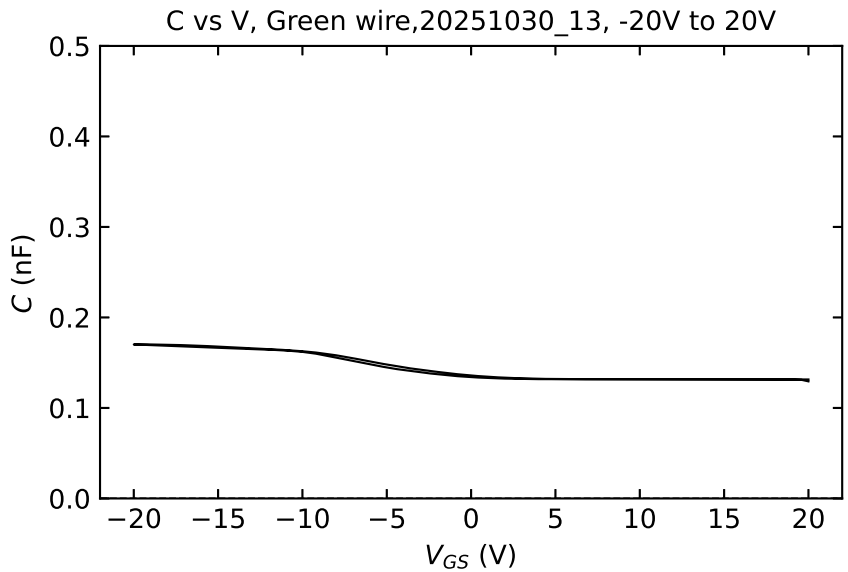



Figure 5: Capacitance curve for PaN sample, green wire contact, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251117001.dat`, **Python:** `20251117001.py`

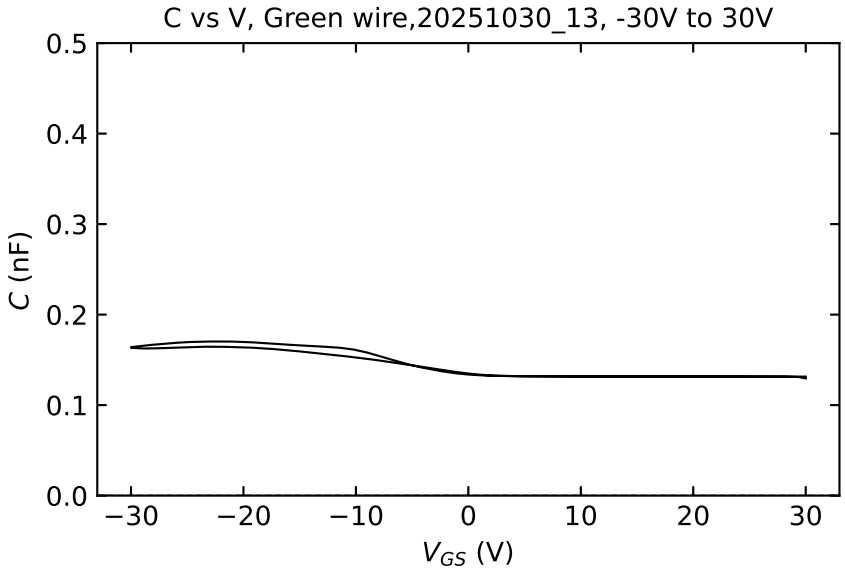



Figure 6: Capacitance curve for PaN sample, green wire contact, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251117002.dat`, **Python:** `20251117002.py`

### Red wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/CapLIA/Red`
- File numbers: `20251117003.dat (20V to -20V)`, `20251117004.dat (30V to -30V)`
- Python: `20251117003.py`, `20251117004.py`

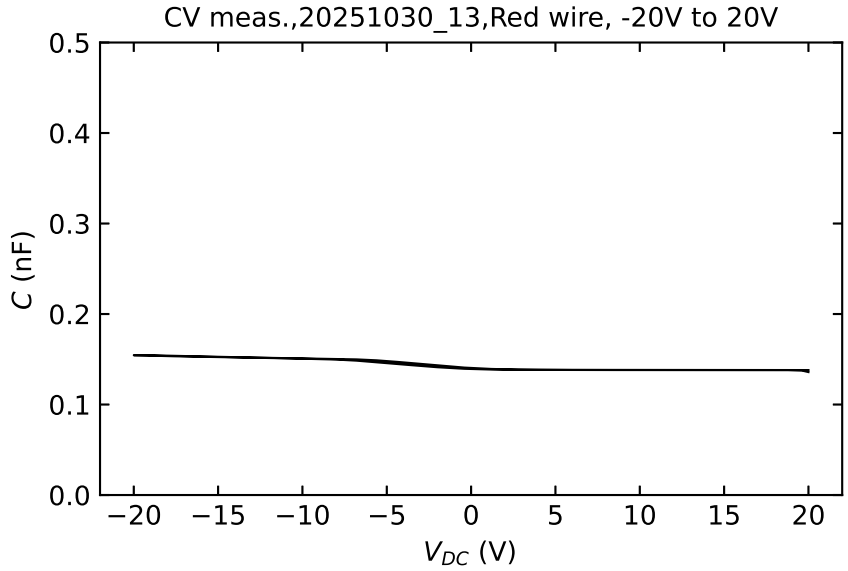



Figure 7: Capacitance curve for PaN sample, red wire contact, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251117003.dat`, **Python:** `20251117003.py`

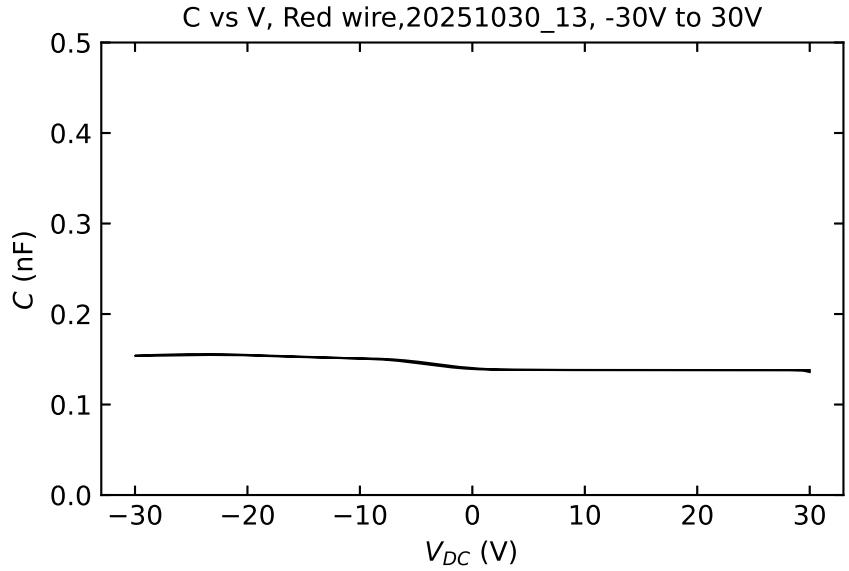



Figure 8: Capacitance curve for PaN sample, red wire contact, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251117004.dat`, **Python:** `20251117004.py`

## $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ Measurements

**Source: Red wire, Drain: Green wire**

- Data directory 20V to -20V: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/I_DS vs V_GS (V_DS)/20V`
- File number: `20251117001.dat`, Python: `20251117001.py`, EPS: `yySec010005`
- Data directory 30V to -30V: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PaN/I_DS vs V_GS (V_DS)/30V`
- File number: `20251117002.dat`, Python: `20251117002.py`, EPS: `yySec010006`

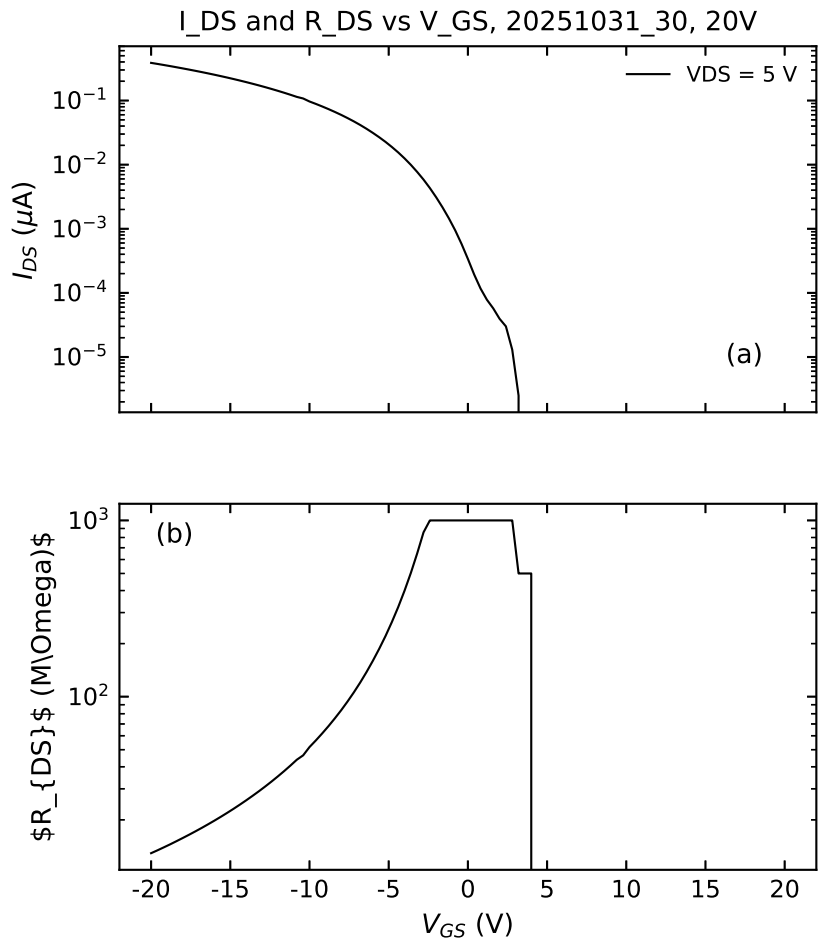



Figure 9: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for PaN sample, $V_{\mathrm{DS}} = 5$V, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251117001.dat`, **Python:** `20251117001.py`

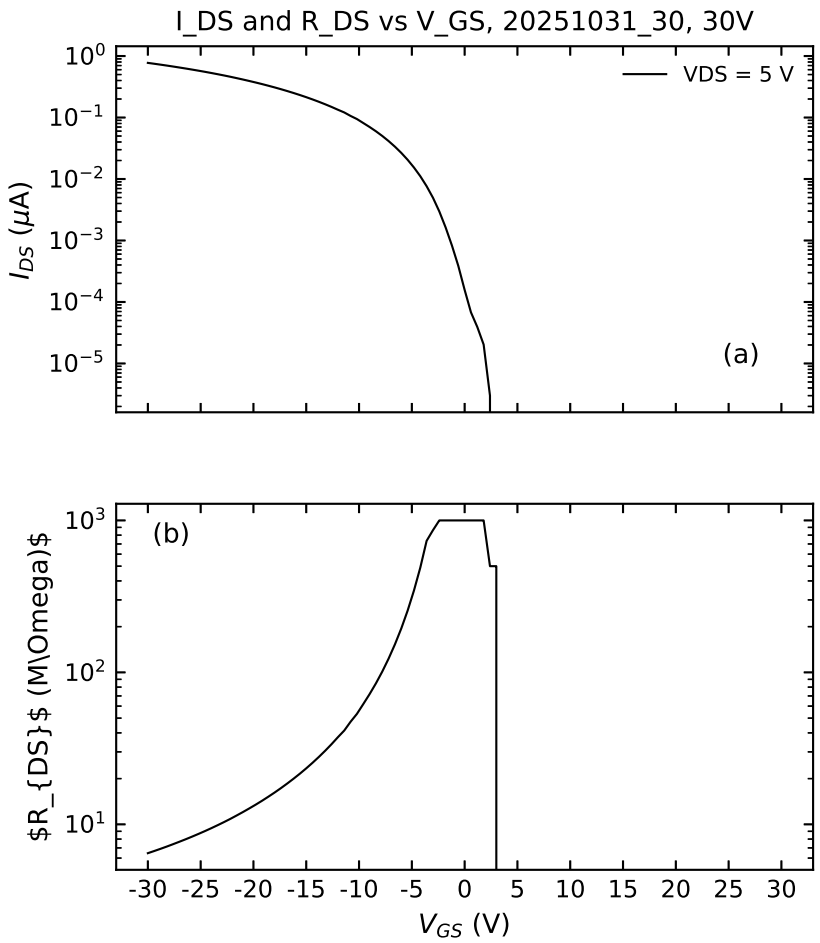



Figure 10: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for PaN sample, $V_{\mathrm{DS}} = 5$V, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251117002.dat`, **Python:** `20251117002.py`

## PS Sample: 20251030_9

### IV Measurements

**Red wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/IV/Red`
- File numbers: `20251117001.dat--20251117006.dat` (VGS = -5V to -30V)
- Python: `2025111701.py`, EPS: `yySec010011`

**Black wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/IV/Black`
- File numbers: `20251117007.dat--20251117012.dat` (VGS = -5V to -30V)
- Python: `2025111702.py`, EPS: `yySec010012`

**Green wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/IV/Green`
- File numbers: `20251117013.dat--20251117018.dat` (VGS = -5V to -30V)

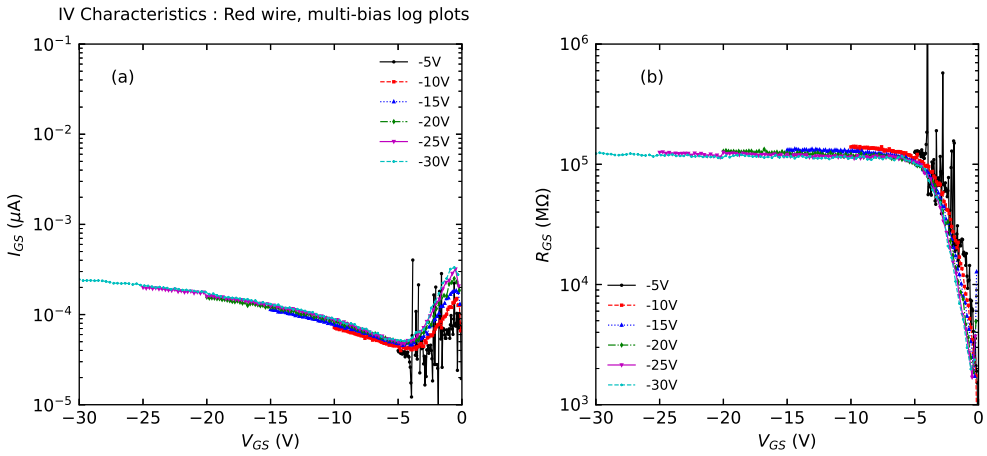



Figure 11: IV curve for PS sample, red wire contact, VGS = -5V to -30V. **Data:** `20251117001.dat--20251117006.dat`, **Python:** `2025111701.py`

- Python: `2025111703.py`, EPS: `yySec010013`

**Yellow wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/IV/Yellow`
- File numbers: `20251117019.dat--20251117024.dat` (VGS = -5V to -30V)
- Python: `2025111704.py`, EPS: `yySec010014`

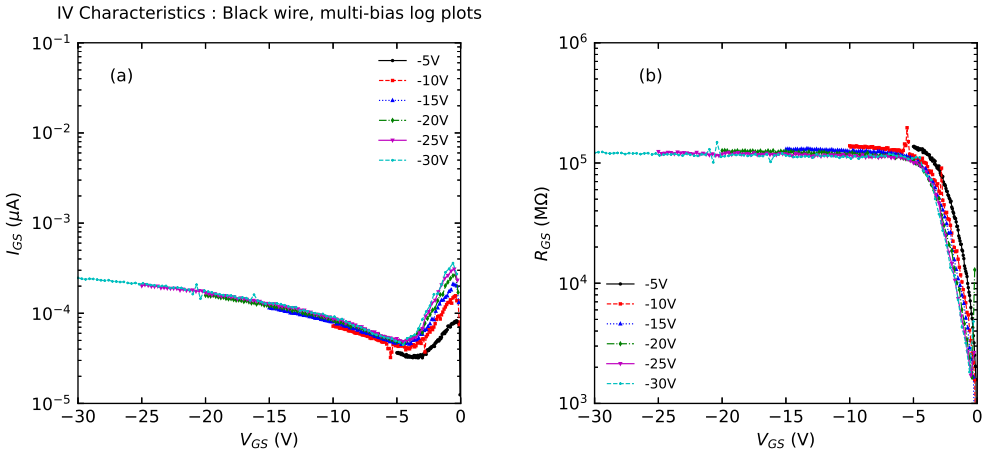



Figure 12: IV curve for PS sample, black wire contact, VGS = -5V to -30V. **Data:** `20251117007.dat--20251117012.dat`, **Python:** `2025111702.py`

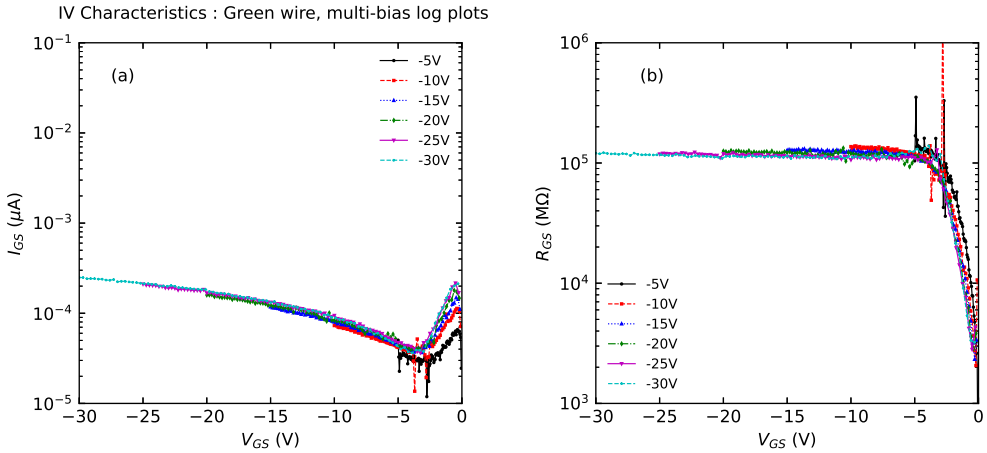



Figure 13: IV curve for PS sample, green wire contact, VGS = -5V to -30V. **Data:** `20251117013.dat--20251117018.dat`, **Python:** `2025111703.py`

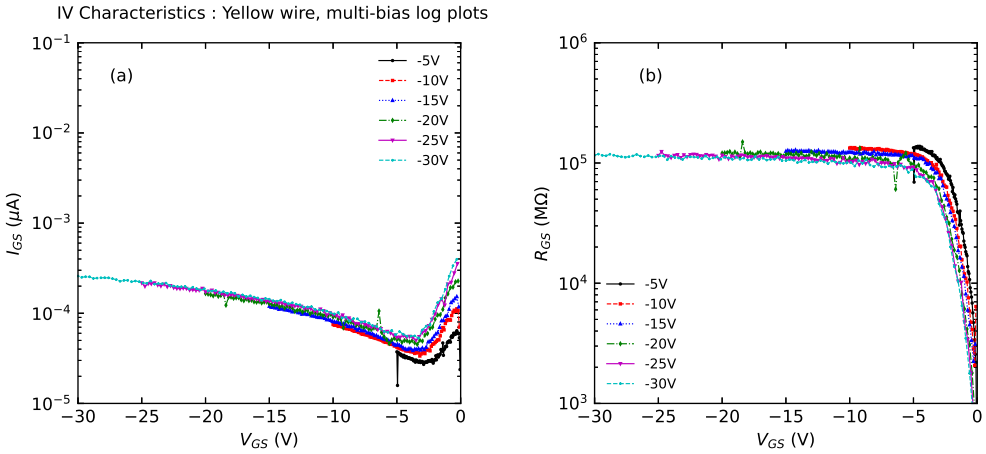



Figure 14: IV curve for PS sample, yellow wire contact, VGS = -5V to -30V. **Data:** `20251117019.dat--20251117024.dat`, **Python:** `2025111704.py`

## Capacitance Measurements (LIA)

### Green wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/CapLIA/Green`
- File numbers: `20251117003.dat (20V to -20V)`, `20251117004.dat (30V to -30V)`
- Python: `20251117003.py`, `20251117004.py`
- EPS: `yySec010011`, `yySec010012`

### Red wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/CapLIA/Red`
- File numbers: `20251117001.dat (20V to -20V)`, `20251117002.dat (30V to -30V)`
- Python: `20251117001.py`, `20251117002.py`
- EPS: `yySec010017`, `yySec010018`

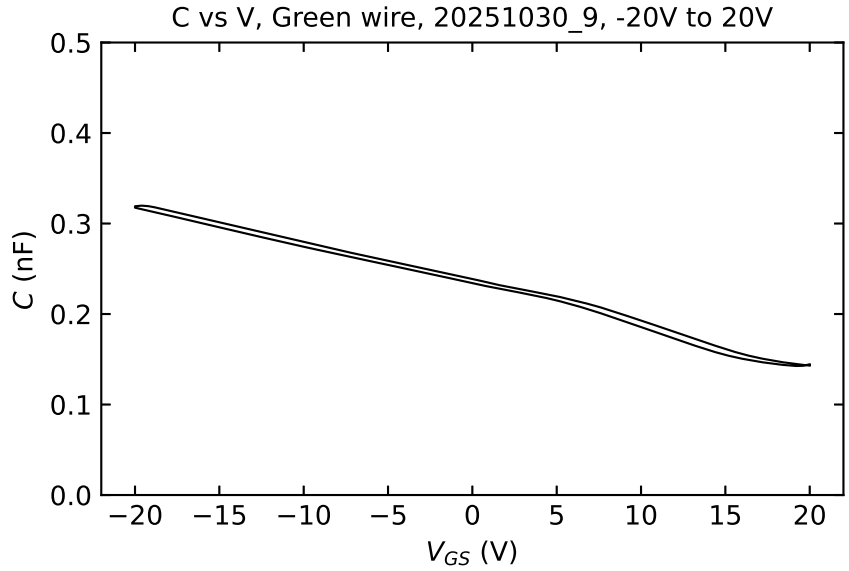



Figure 15: Capacitance curve for PS sample, green wire contact, VGS = 20V to -20V. **Data:** `20251117003.dat`, **Python:** `20251117003.py`

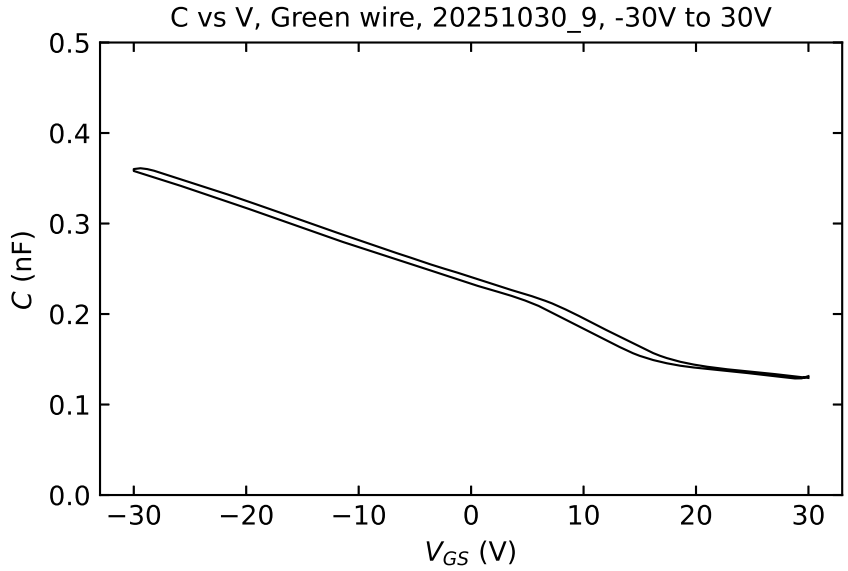



Figure 16: Capacitance curve for PS sample, green wire contact, VGS = 30V to -30V. **Data:** `20251117004.dat`, **Python:** `20251117004.py`

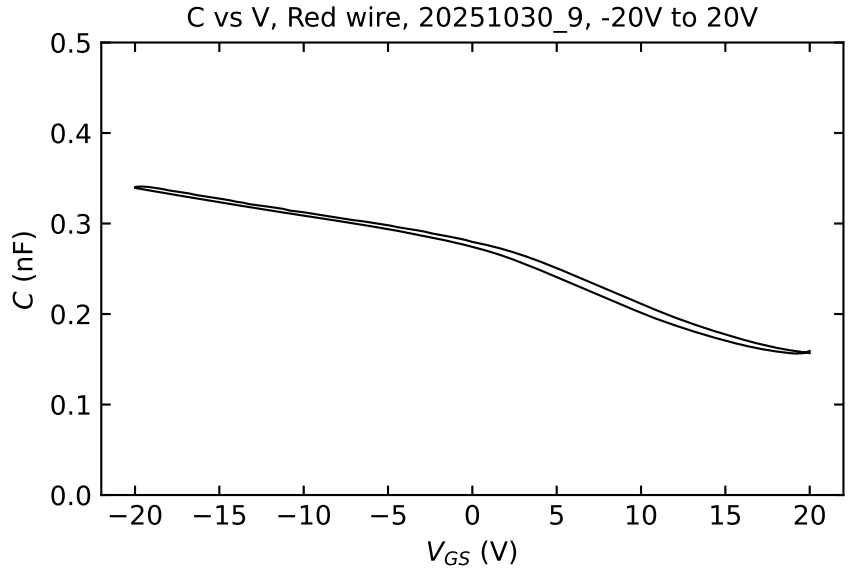



Figure 17: Capacitance curve for PS sample, red wire contact, VGS = 20V to -20V. **Data:** `20251117001.dat`, **Python:** `20251117001.py`

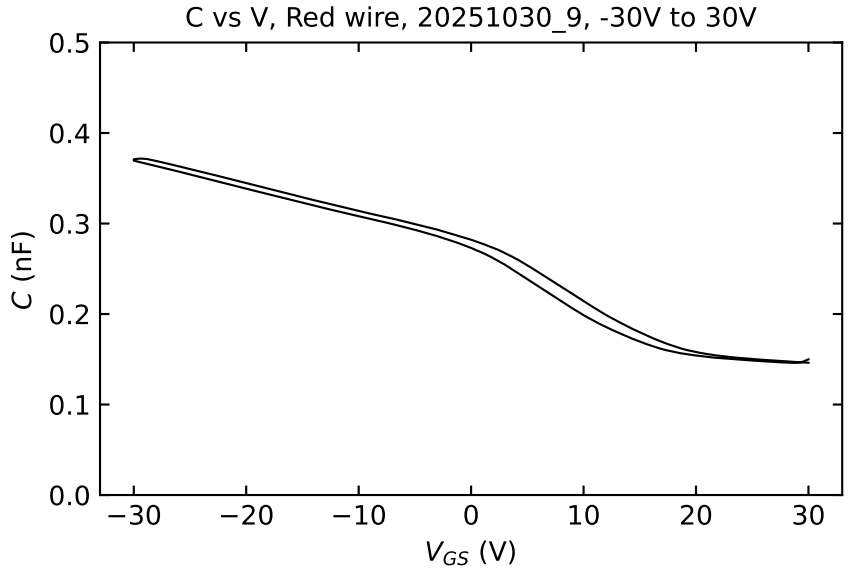



Figure 18: Capacitance curve for PS sample, red wire contact, VGS = 30V to -30V. **Data:** `20251117002.dat`, **Python:** `20251117002.py`

## $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ Measurements

**Source: Red wire, Drain: Green wire**

- Data directory 20V to -20V: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/I_DS vs V_GS (V_DS)/20V`
- File numbers: `20251117001.dat--20251117005.dat`, Python: `20251117.py`, EPS: `yySec010015`
- Data directory 30V to -30V: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/PS/I_DS vs V_GS (V_DS)/30V`
- File numbers: `20251117001.dat--20251117005.dat`, Python: `20251117.py`, EPS: `yySec010016`

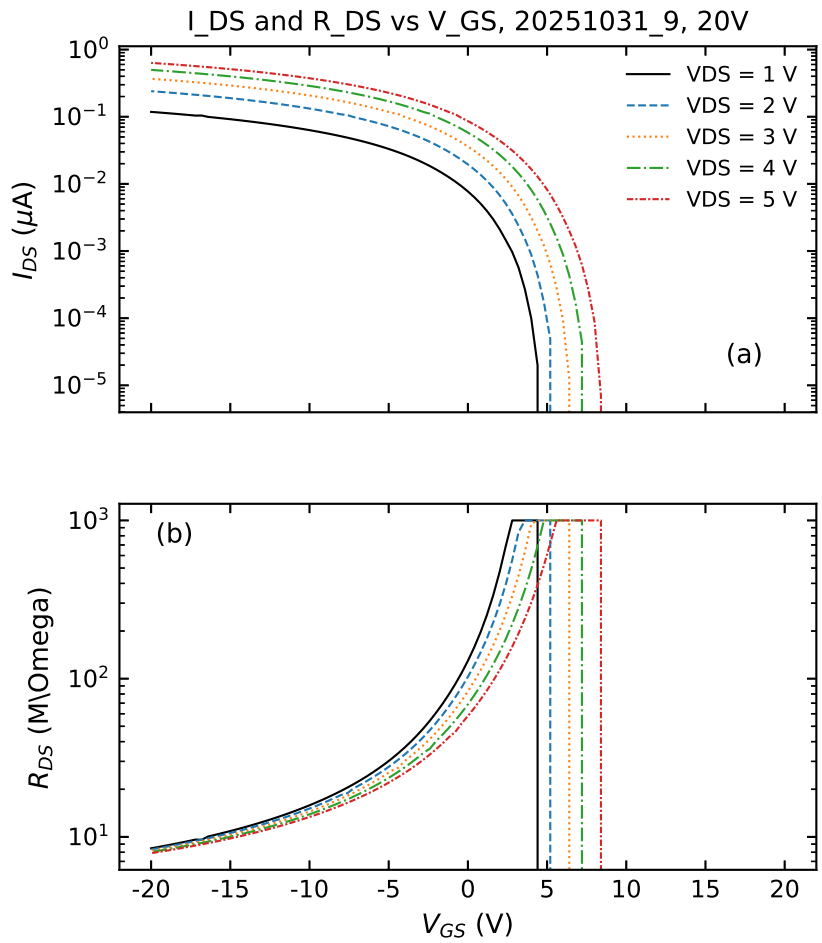



Figure 19: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for PS sample, $V_{\mathrm{DS}} = 1$V to 5V, VGS = 20V to -20V. **Data:** `20251117001.dat--20251117005.dat`, **Python:** `20251117.py`

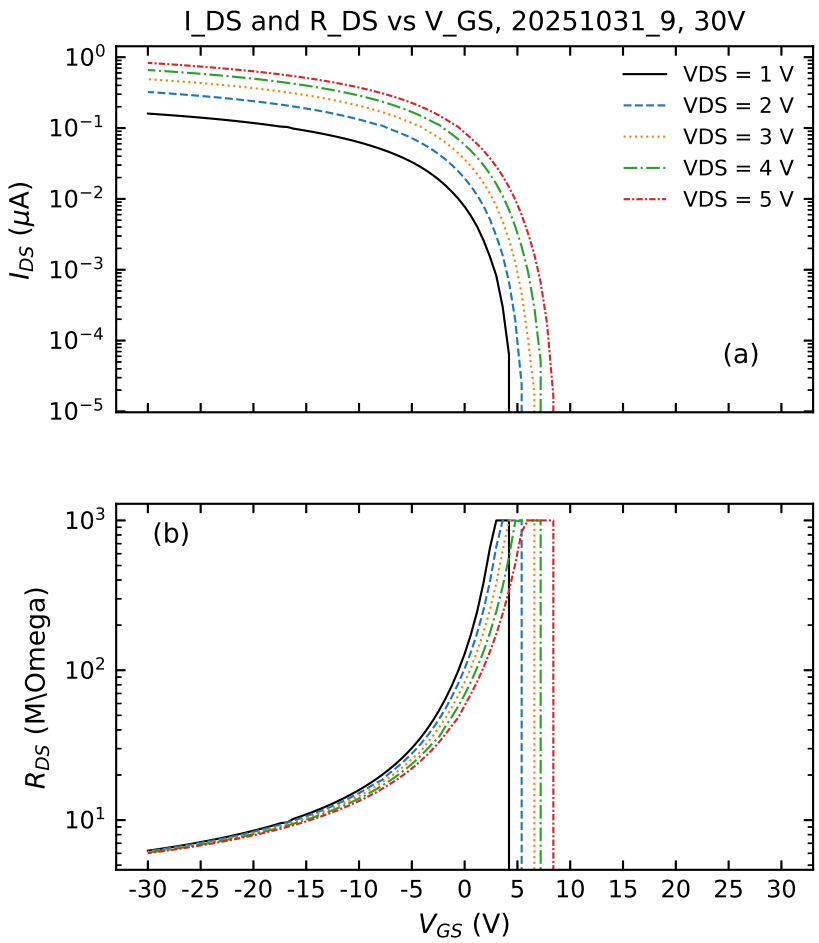



Figure 20: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for PS sample, $V_{\mathrm{DS}} =$ 1V to 5V, VGS = 30V to -30V. **Data:** `20251117001.dat--20251117005.dat`, **Python:** `20251117.py`

## Trip-C12 Sample: 20251030_15

### IV Measurements

### Black wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/IV/B`
- File numbers: `20251118001.dat--20251118006.dat` (VGS = -5V to -30V)
- Python: `2025111801.py`, EPS: `yySec010021`

### Red wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/IV/Red`
- File numbers: `20251118007.dat--20251118012.dat` (VGS = -5V to -30V)
- Python: `2025111802.py`, EPS: `yySec01022`

### Green wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/IV/Green`
- File numbers: `20251118013.dat--20251118018.dat` (VGS = -5V to -30V)

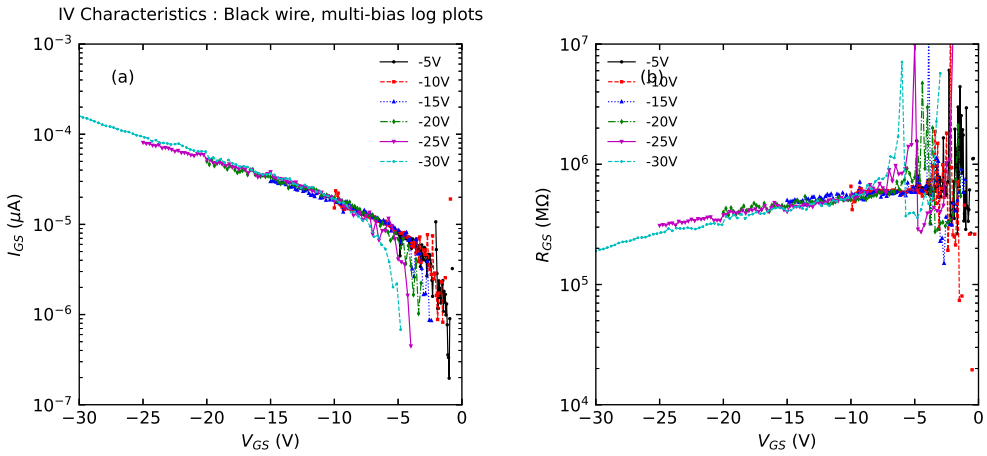



Figure 21: IV curve for Trip-C12 sample, black wire contact, VGS = -5V to -30V. **Data:** `20251118001.dat--20251118006.dat`, **Python:** `2025111801.py`

- Python: `2025111803.py`, EPS: `yySec010023`

### Yellow wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/IV/Yello`
- File numbers: `20251118019.dat--20251118024.dat` (VGS = -5V to -30V)
- Python: `2025111804.py`, EPS: `yySec010024`

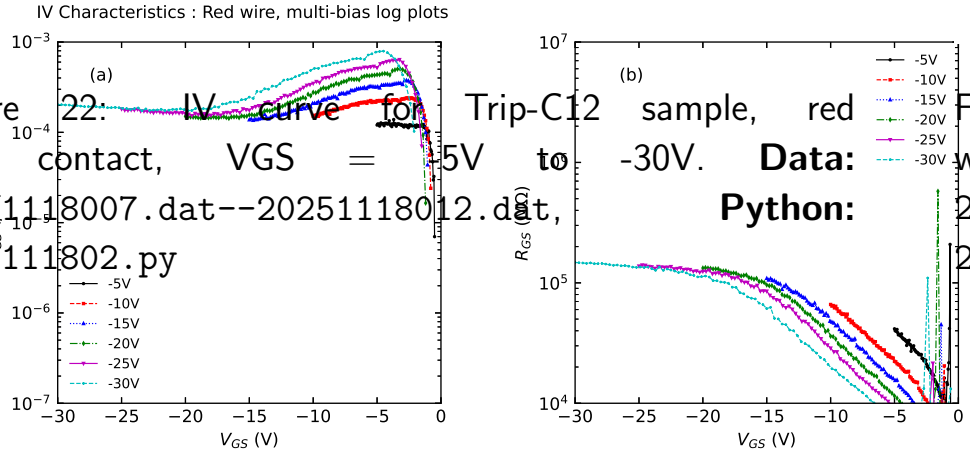



Figure 22: IV curve for Trip-C12 sample, red wire contact, VGS = -5V to -30V. **Data:** `20251118007.dat--20251118012.dat`, **Python:** `2025111802.py`

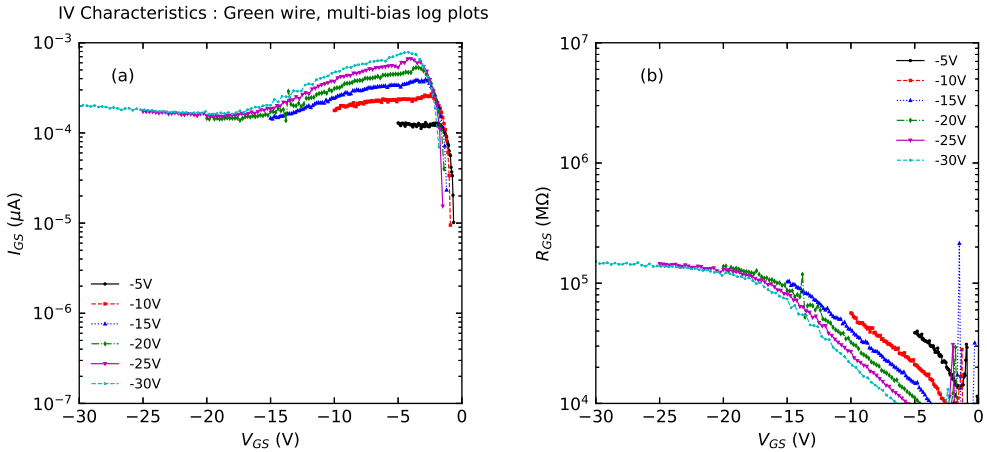



Figure 23: IV curve for Trip-C12 sample, green wire contact, VGS = -5V to -30V. **Data:** `20251118013.dat--20251118018.dat`, **Python:** `2025111803.py`

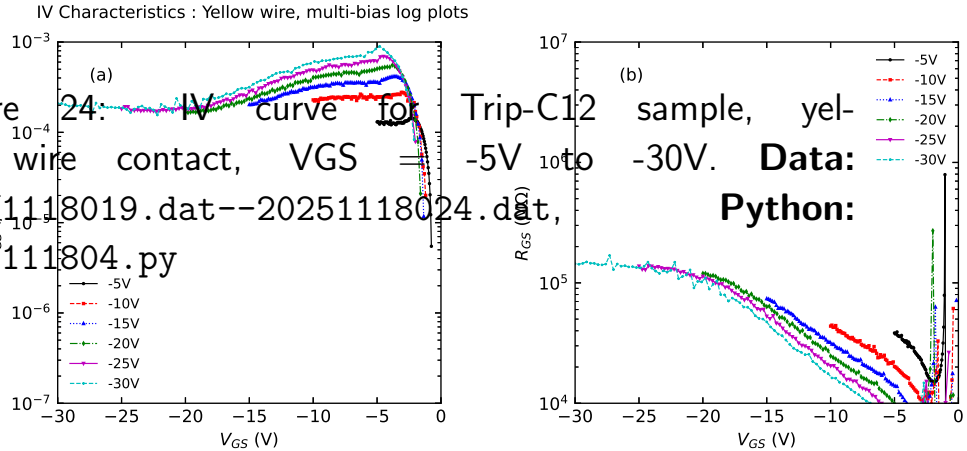



Figure 24: IV curve for Trip-C12 sample, yellow wire contact, VGS = -5V to -30V. **Data:** `20251118019.dat--20251118024.dat`, **Python:** `2025111804.py`

## Capacitance Measurements (LIA)

### Red wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/CapLIA/Red`
- File numbers: `20251118001.dat` (20V to -20V), `20251118002.dat` (30V to -30V)
- Python: `20251118001.py`, `20251118002.py`, EPS: `yySec010025`, `yySec010026`

### Green wire contact:

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/CapLI`
- File numbers: `20251118003.dat` (20V to -20V), `20251118004.dat` (30V to -30V)
- Python: `20251118003.py`, `20251118004.py`, EPS: `yySec010027`, `yySec010028`

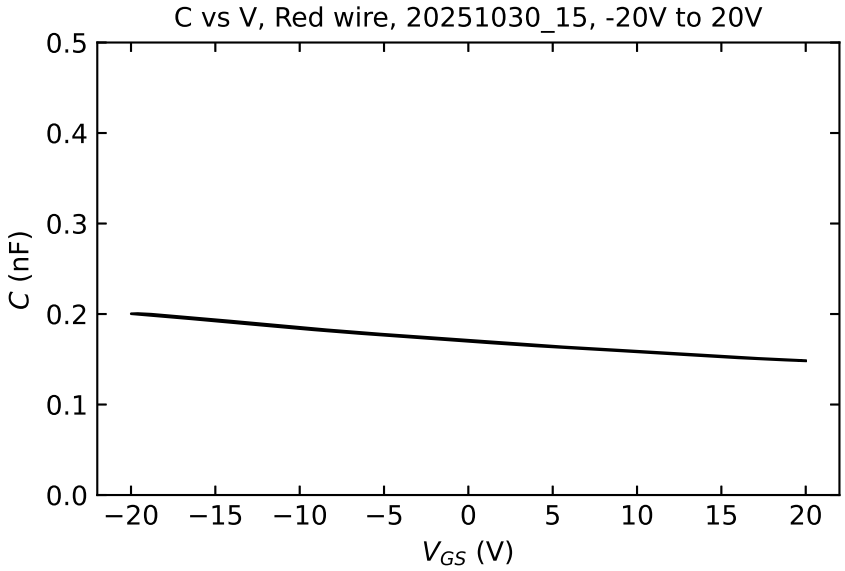



Figure 25: Capacitance curve for Trip-C12 sample, red wire contact, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251118001.dat`, **Python:** `20251118001.py`

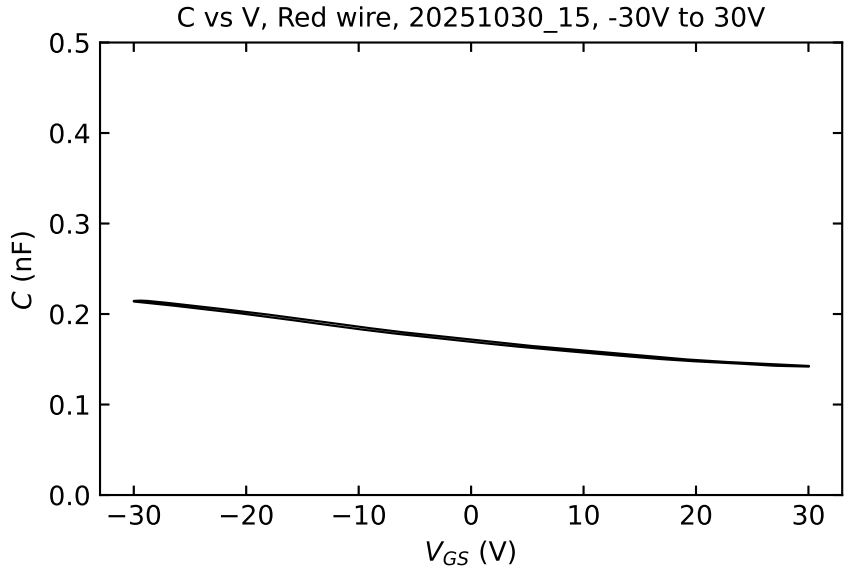



Figure 26: Capacitance curve for Trip-C12 sample, red wire contact, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251118002.dat`, **Python:** `20251118002.py`

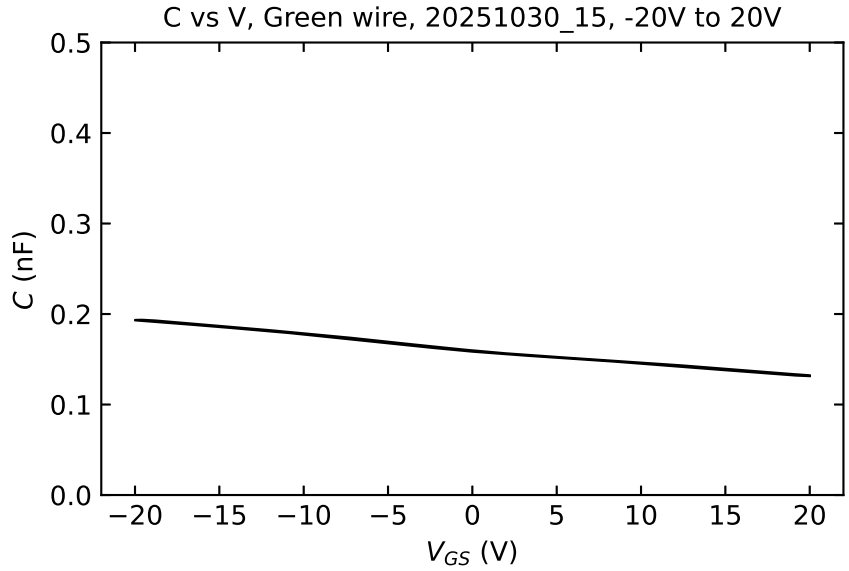



Figure 27: Capacitance curve for Trip-C12 sample, green wire contact, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251118003.dat`, **Python:** `20251118003.py`

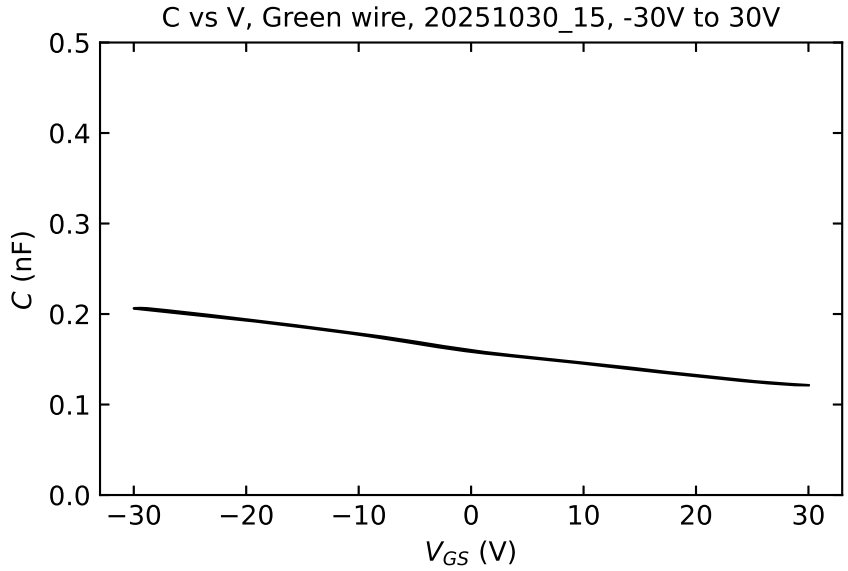
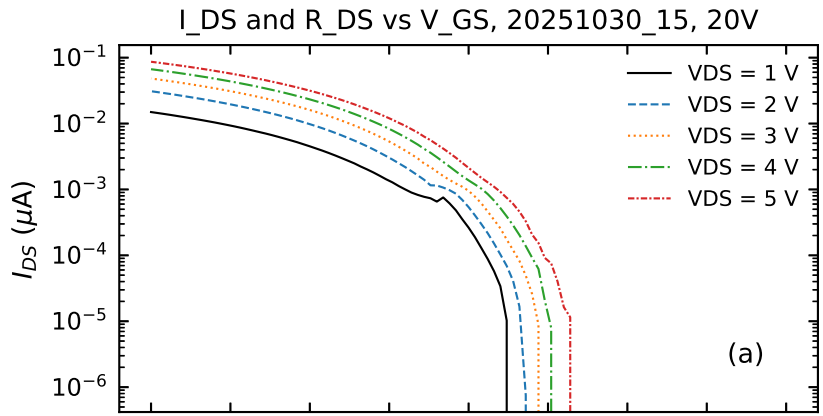



Figure 28: Capacitance curve for Trip-C12 sample, green wire contact, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251118004.dat`, **Python:** `20251118004.py`

## $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ Measurements

**Source: Red wire, Drain: Green wire**

- Data directory 20V to -20V: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/Trip-C_12/I_DS vs V_GS (V_DS)/20V`
- File numbers: `20251118001.dat--20251118005.dat`, Python: `20251118.py`, EPS: `yySec010029`
- Data directory 30V 30V: `Q:/THz-Lal Data/AllMeasuredData/2025/20251 vs V_GS (V_DS)/30V`
- File numbers: `20251118001.dat--202` Python: `20251118.py`, EPS: `yySec01`



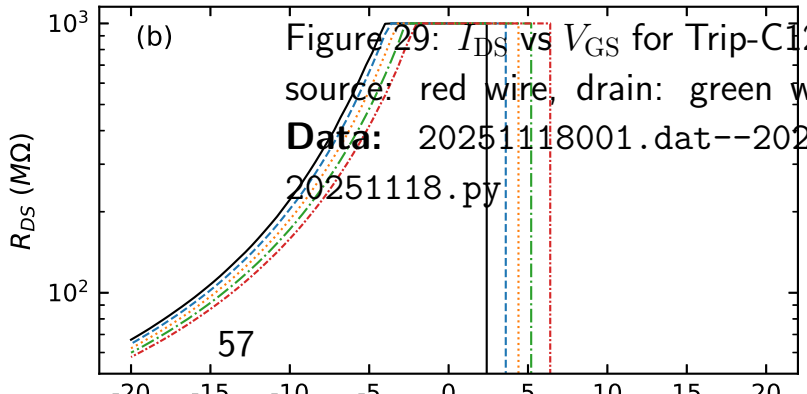




Figure 29: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for Trip-C12 sample, $V_{\mathrm{DS}} = 1$V to 5V, source: red wire, drain: green wire, $V_{\mathrm{GS}} = 20$V to -20V. **Data:** `20251118001.dat--20251118005.dat`, **Python:** `20251118.py`

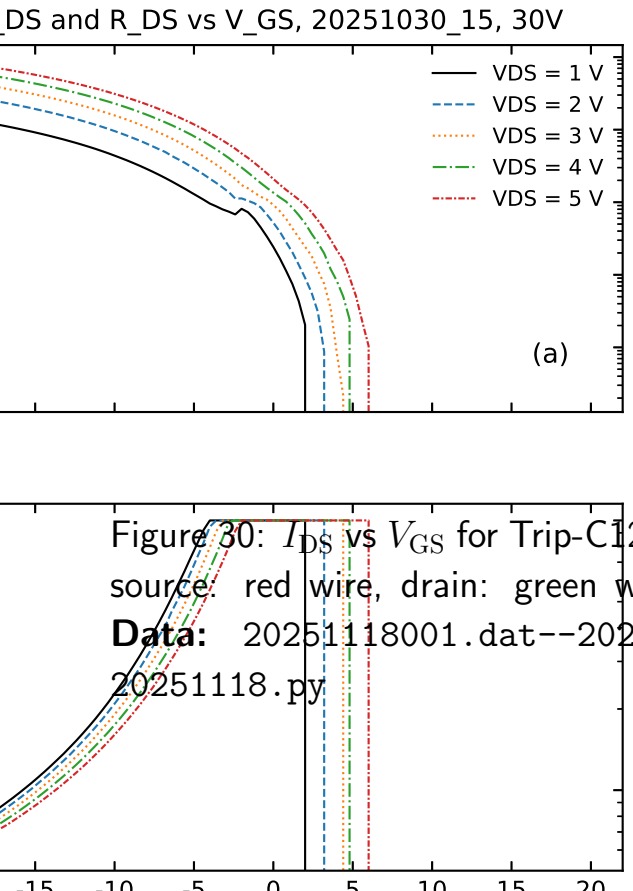



Figure 30: $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ for Trip-C12 sample, $V_{\mathrm{DS}} = 1$V to 5V, source: red wire, drain: green wire, $V_{\mathrm{GS}} = 30$V to -30V. **Data:** `20251118001.dat--20251118005.dat`, **Python:** `20251118.py`

## Comparison Between Samples

### Capacitance ($C$) vs $V_{\mathrm{GS}}$ Comparison

**Green wire contact, 20V to -20V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/C vs V_GS/-20V to +20V/Green wire`
- File numbers: PaN: `2025111701.dat`, PS: `2025111702.dat`, Trip-C12: `2025111803.dat`
- Python: `2025111701.py`, EPS: `yySec010031`

**Red wire contact, 20V to -20V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/C vs V_GS/-20V to +20V/Red wire`
- File numbers: PaN: `2025111701.dat`, PS: `2025111702.dat`, Trip-C12: `2025111803.dat`
- Python: `2025111702.py`, EPS: `yySec010032`

**Green wire contact, 30V to -30V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/C vs V_GS/-30V to +30V/Green wire`

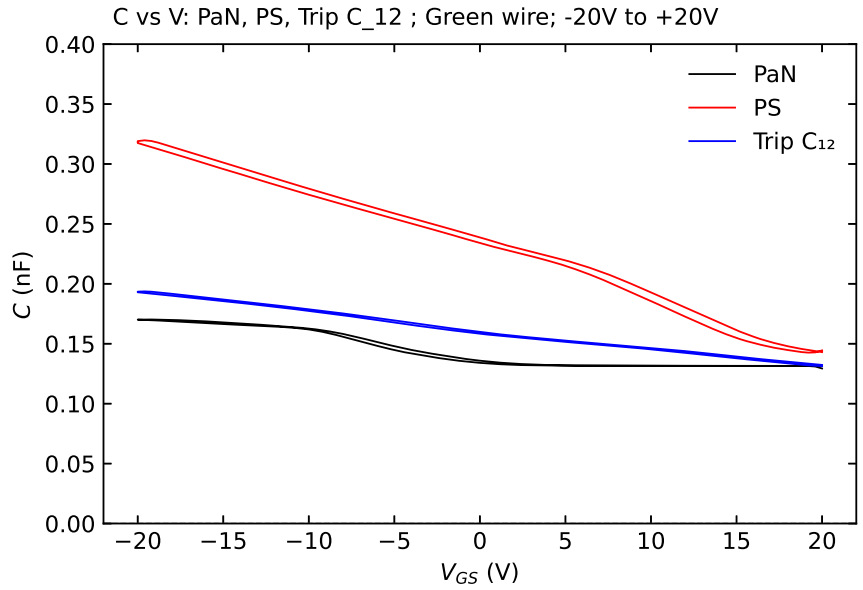



Figure 31: $C$ vs $V_{\mathrm{GS}}$ comparison (green wire, 20V to -20V) for PaN, PS, and Trip-C12 samples.

- File numbers: PaN: `2025111701.dat`, PS: `2025111702.dat`, Trip-C12: `2025111803.dat`
- Python: `2025111703.py`, EPS: `yySec01033`

**Red wire contact, 30V to -30V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/C vs V_GS/-30V to +30V/Red wire`
- File numbers: PaN: `2025111701.dat`, PS:

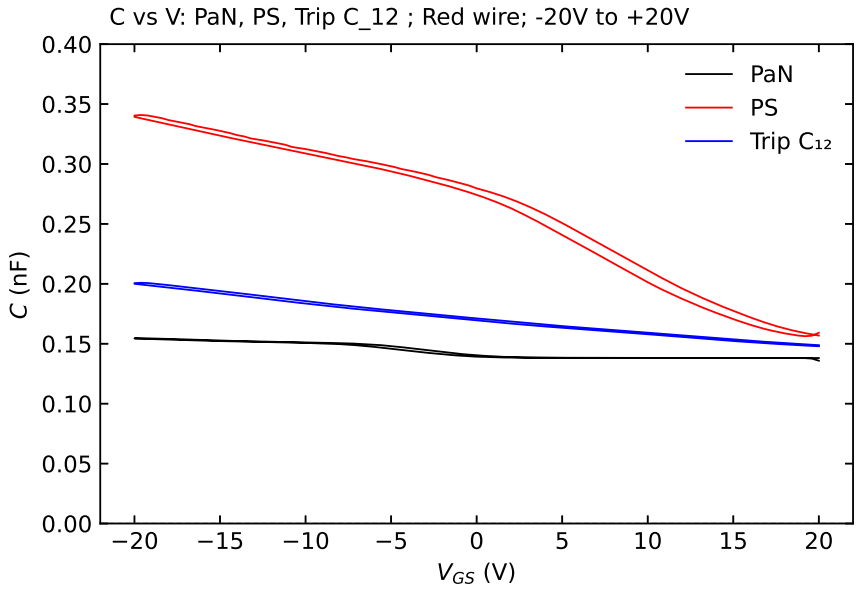



Figure 32: $C$ vs $V_{\mathrm{GS}}$ comparison (red wire, 20V to -20V) for PaN, PS, and Trip-C12 samples.

`2025111702.dat`, Trip-C12: `2025111803.dat`

- Python: `2025111703.py`, EPS: `yySec010034`

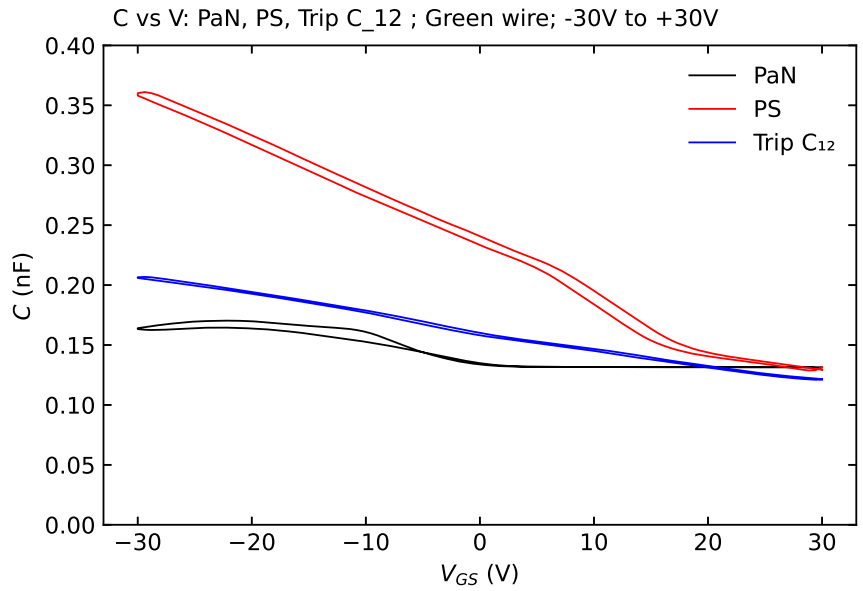



Figure 33: $C$ vs $V_{\mathrm{GS}}$ comparison (green wire, 30V to -30V) for PaN, PS, and Trip-C12 samples.

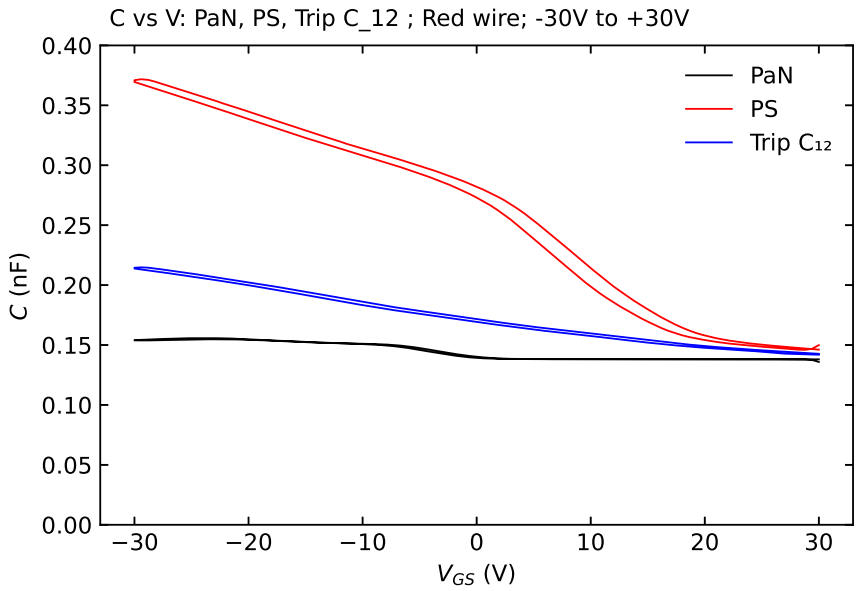



Figure 34: $C$ vs $V_{\mathrm{GS}}$ comparison (red wire, 30V to -30V) for PaN, PS, and Trip-C12 samples.

### $I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$ Comparison Between Samples

**20V to -20V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/I_DS vs V_GS/-20V to +20V`
- File numbers: PaN: `2025111701.dat`, PS: `2025111702.dat`, Trip-C12: `2025111803.dat`
- Python: `20251117.py`, EPS: `yySec010035`

**30V to -30V:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251117/I_DS vs V_GS/-30V to +30V`
- File numbers: PaN: `2025111701.dat`. PS: `2025111702.dat`, Trip-C12: `2025111`
- Python: `20251117.py`, EPS: `yySec01`

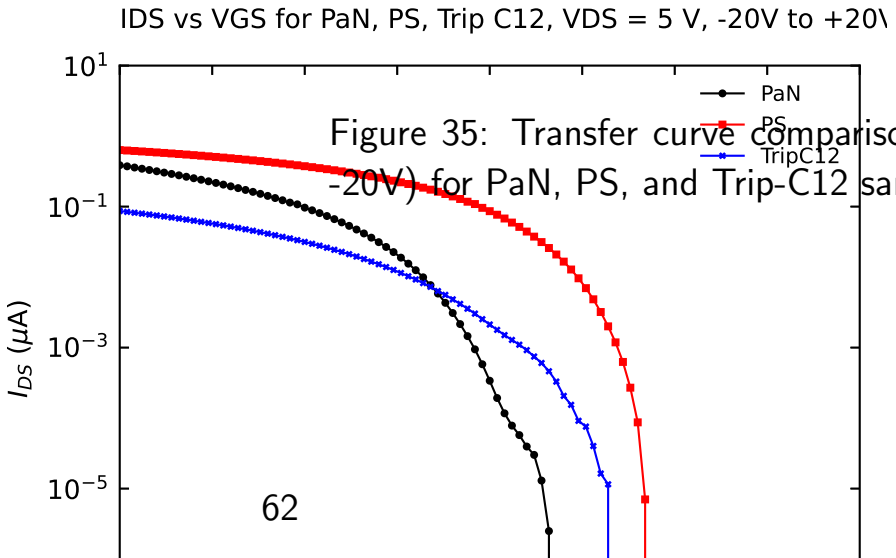



Figure 35: Transfer curve comparison ($I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$, 20V to -20V) for PaN, PS, and Trip-C12 samples.

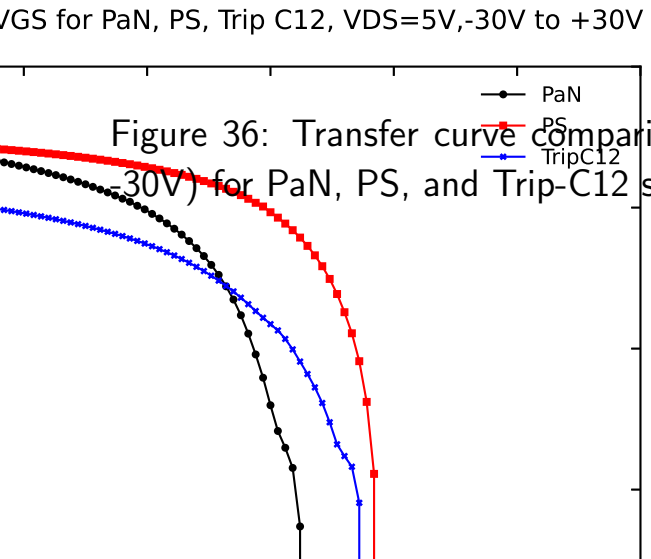



Figure 36: Transfer curve comparison ($I_{\mathrm{DS}}$ vs $V_{\mathrm{GS}}$, 30V to -30V) for PaN, PS, and Trip-C12 samples.

# DLTS measurements (PS+Ph-BTBT-C8)

**Measurement details:** DLTS measurements on sample 20251030_9_PS+Ph-BTBT-C8 with negative gate voltages only ($-30$ V and $-15$ V) on four different contacts. Data was used to calculate $n_{2D}$ by two methods.

**Black wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251128/Different Contacts/Neg/Black wire`
- File numbers: $-30$ V: `20251128001.dat`, $-15$ V: `20251128005.dat`
- Python: $-30$ V: `20251128001.py`, $-15$ V: `20251128005.py`
- EPS: $-30$ V: `yySec010037.eps`, $-15$ V: `yySec010038.eps`

**Red wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251128/Different Contacts/Neg/Red wire`
- File numbers: $-30$ V: `20251128002.dat`, $-15$ V: `20251128006.dat`

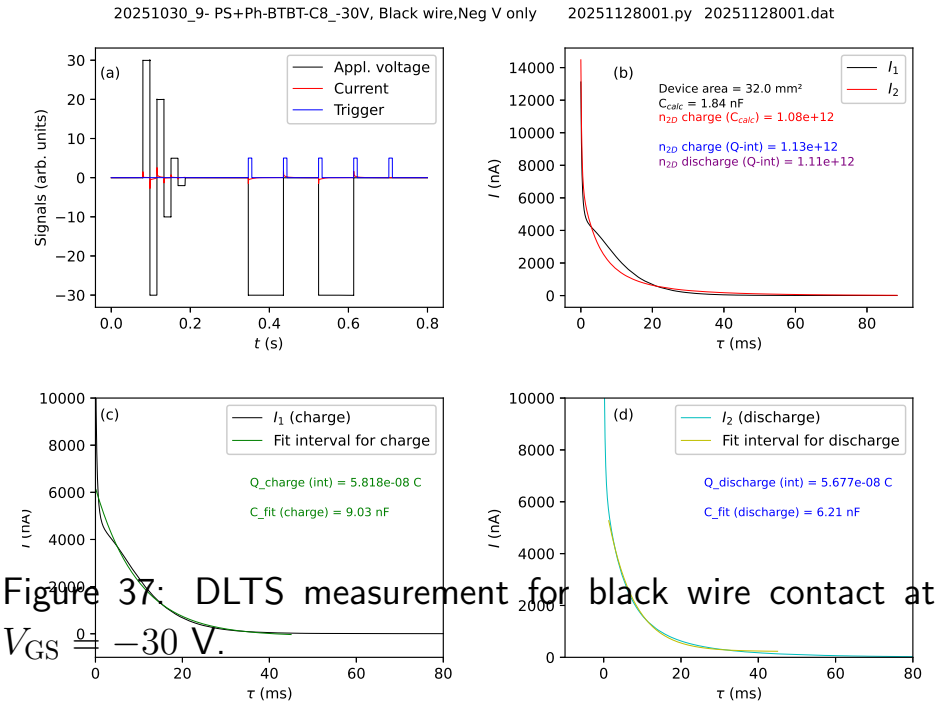



Figure 37: DLTS measurement for black wire contact at $V_{GS} = -30$ V.

- Python: $-30$ V: `20251128002.py`, $-15$ V: `20251128006.py`
- EPS: $-30$ V: `yySec010039.eps`, $-15$ V: `yySec010040.eps`

**Green wire contact:**

- Data directory: `Q:/THz-Lab/Measurement`

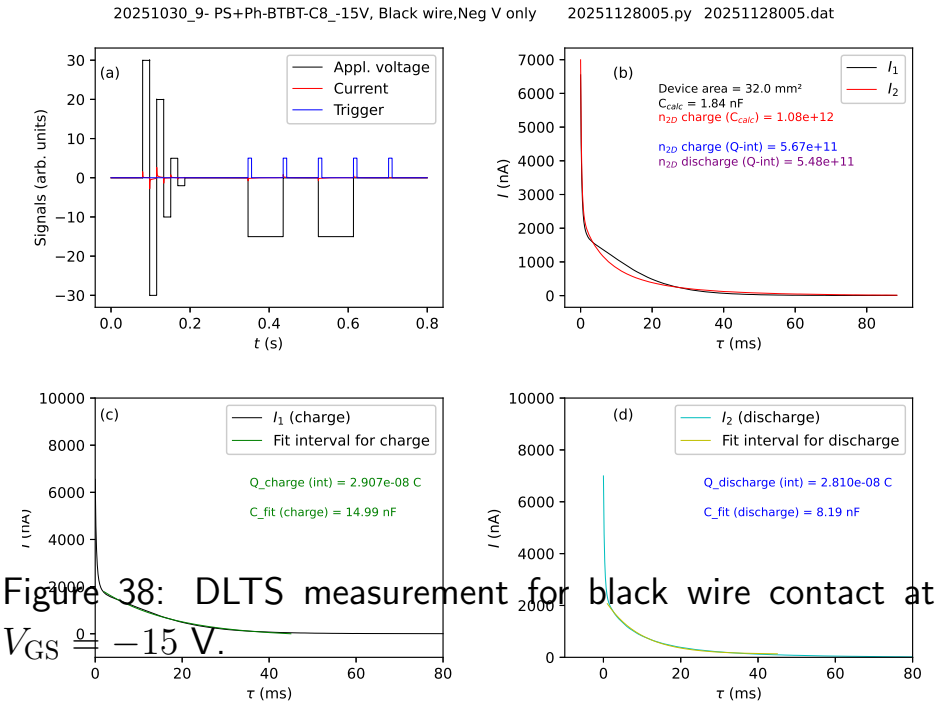



Figure 38: DLTS measurement for black wire contact at $V_{\mathrm{GS}} = -15$ V.

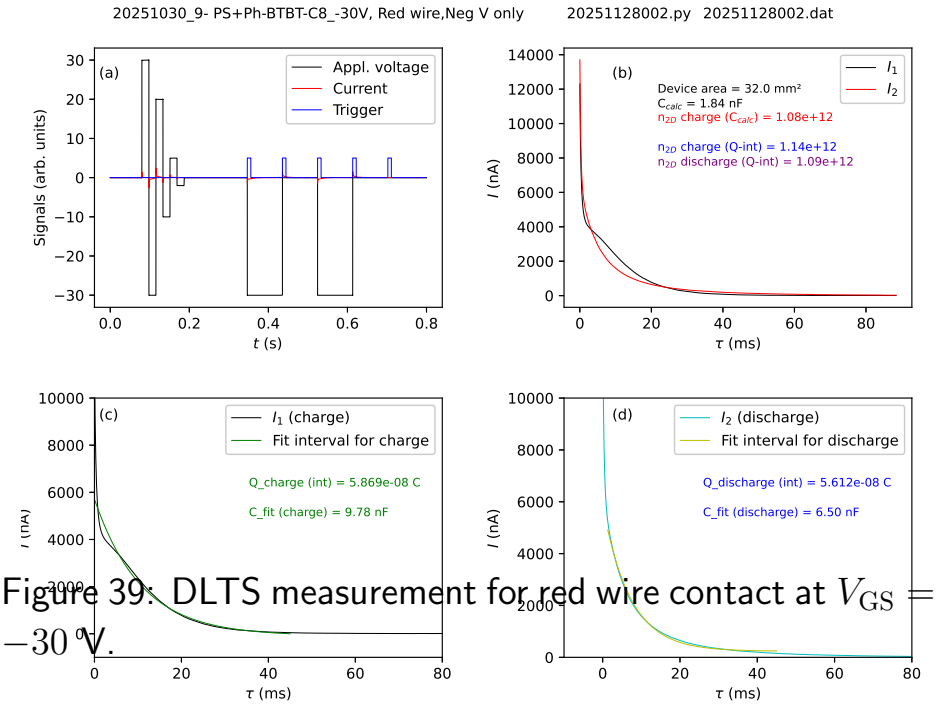



Figure 39: DLTS measurement for red wire contact at $V_{\mathrm{GS}} = -30$ V.

`Data/AllMeasuredData/2025/20251128/Different Contacts/Neg/Green wire`

- File numbers: $-30$ V: `20251128003.dat`, $-15$ V: `20251128007.dat`
- Python: $-30$ V: `20251128003.py`, $-15$ V: `20251128007.py`
- EPS: $-30$ V: `yySec010041.eps`, $-15$ V: `yySec010042.eps`

**Yellow wire contact:**

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251128/Different Contacts/Neg/Yellow wire`
- File numbers: $-30$ V: `20251128004.dat`, $-15$ V: `20251128008.dat`

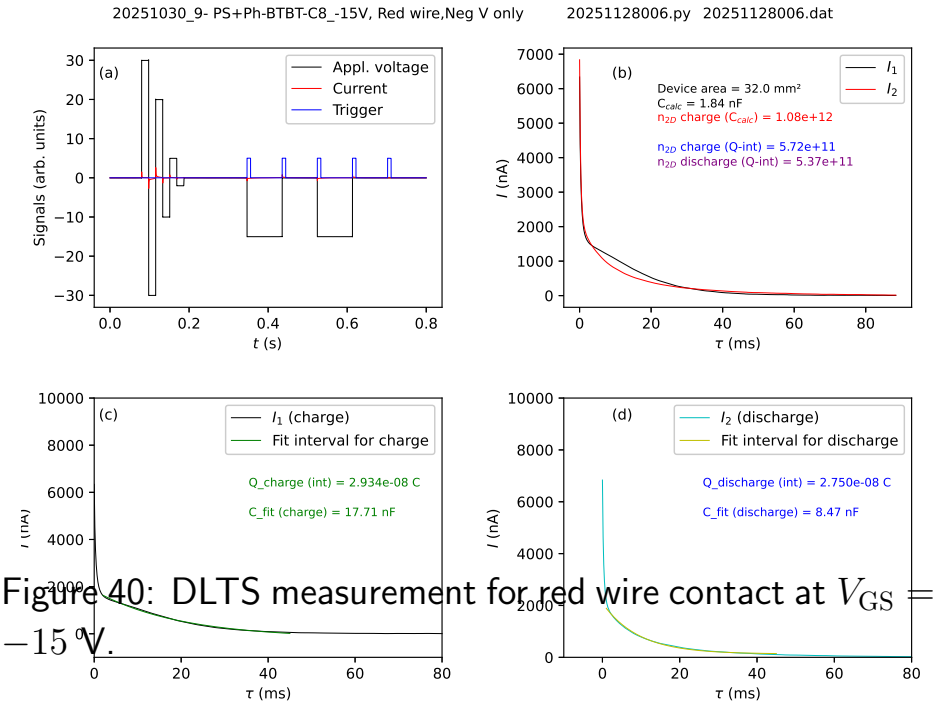



Figure 40: DLTS measurement for red wire contact at $V_{\mathrm{GS}} = -15$ V.

- Python: $-30$ V: `20251128004.py`, $-15$ V: `20251128008.py`
- EPS: $-30$ V: `yySec010043.eps`, $-15$ V: `yySec010044.eps`

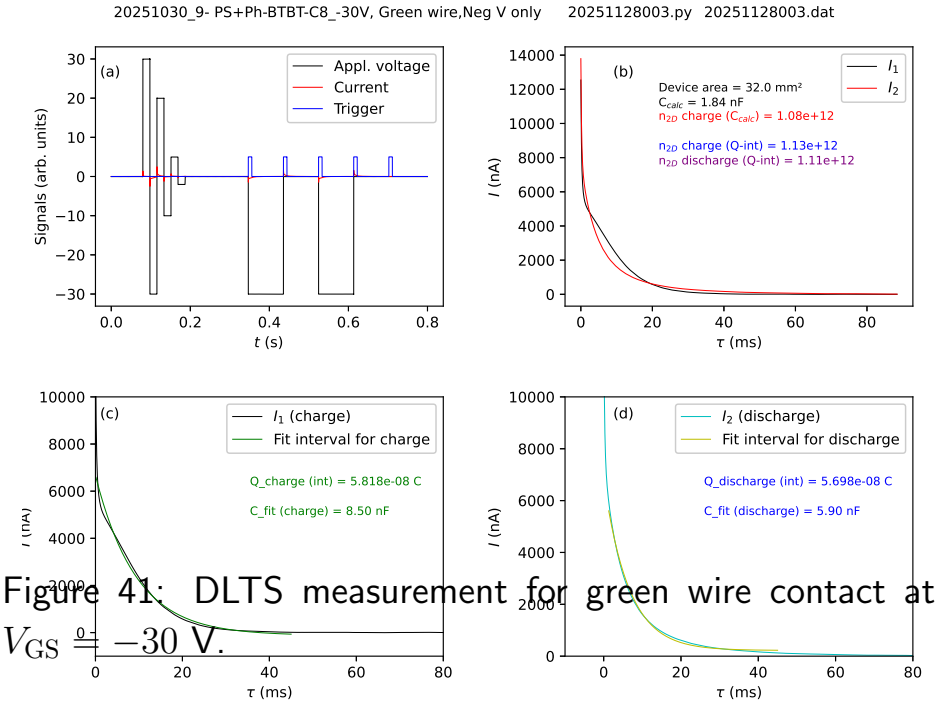



Figure 41: DLTS measurement for green wire contact at $V_{\mathrm{GS}} = -30$ V.

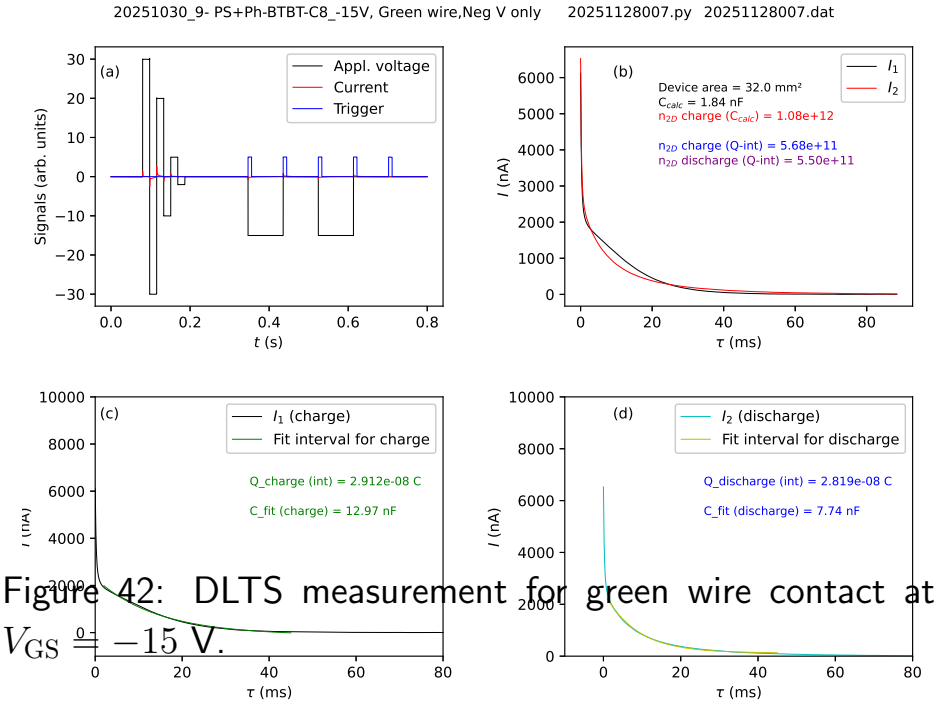



Figure 42: DLTS measurement for green wire contact at $V_{\mathrm{GS}} = -15$ V.

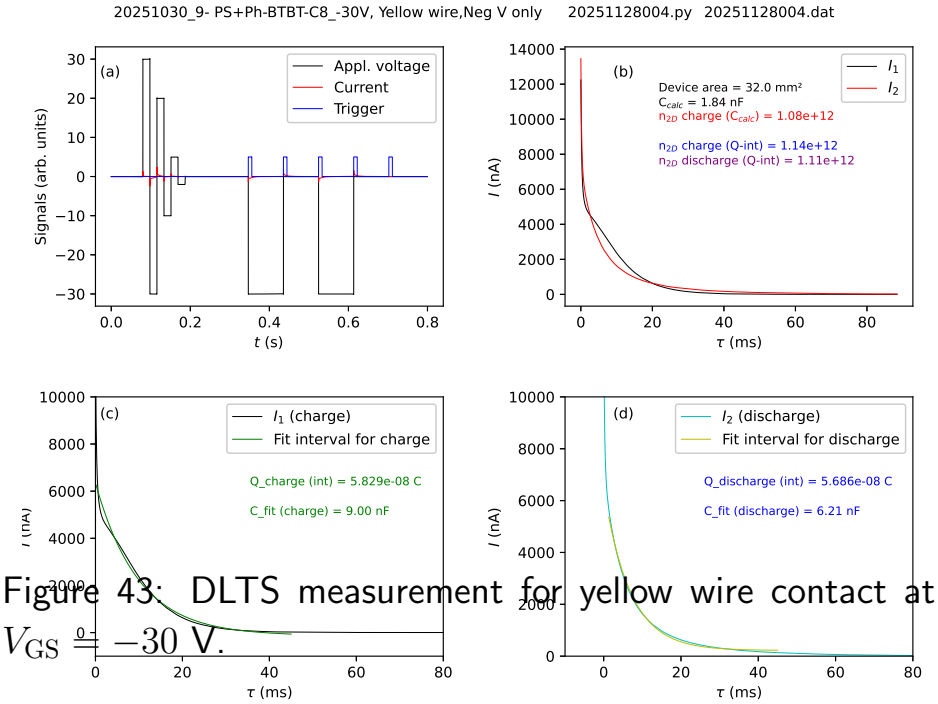



Figure 43: DLTS measurement for yellow wire contact at $V_{\mathrm{GS}} = -30$ V.

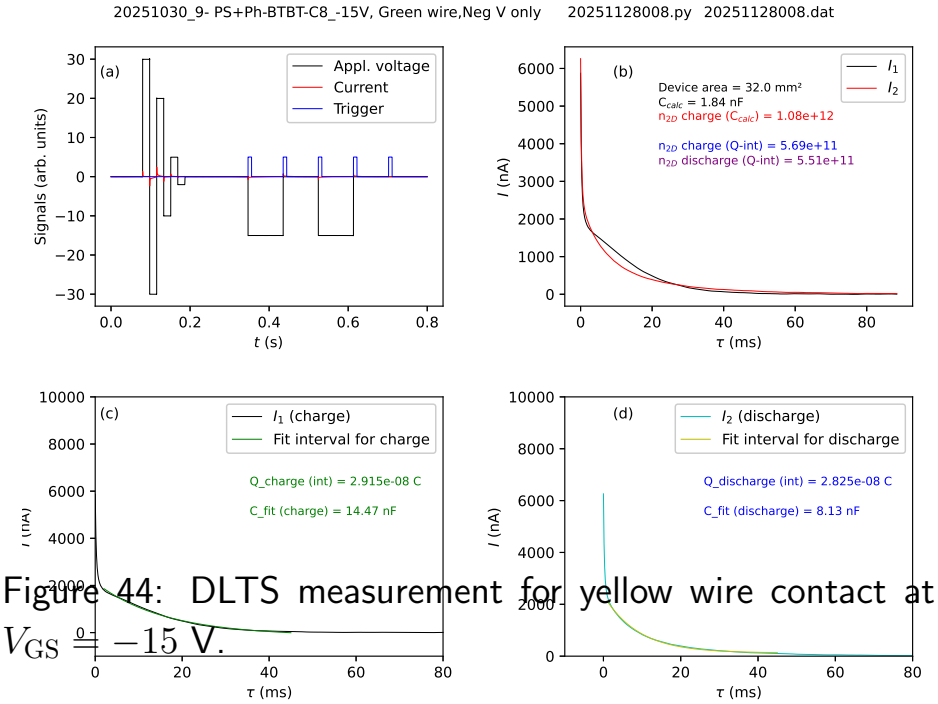



Figure 44: DLTS measurement for yellow wire contact at $V_{\mathrm{GS}} = -15$ V.

## DLTS Sweep Across +6 V to -30 V (Black Wire Contact)

**Measurement details:** Gate sweep from $+6$ V to $-30$ V in steps of 3 V for black wire contact. Data used to calculate $n_{2\mathrm{D}}$.

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251128/Vg_Neg`
- File numbers: 6 V: `20251128013.dat`, 3 V: `20251128012.dat`, 0 V: `20251128011.dat`, -3 V: `20251128001.dat`, ..., -30 V: `20251128010.dat`
- Python: `20251128_4.py`
- EPS: `yySec010045.eps`

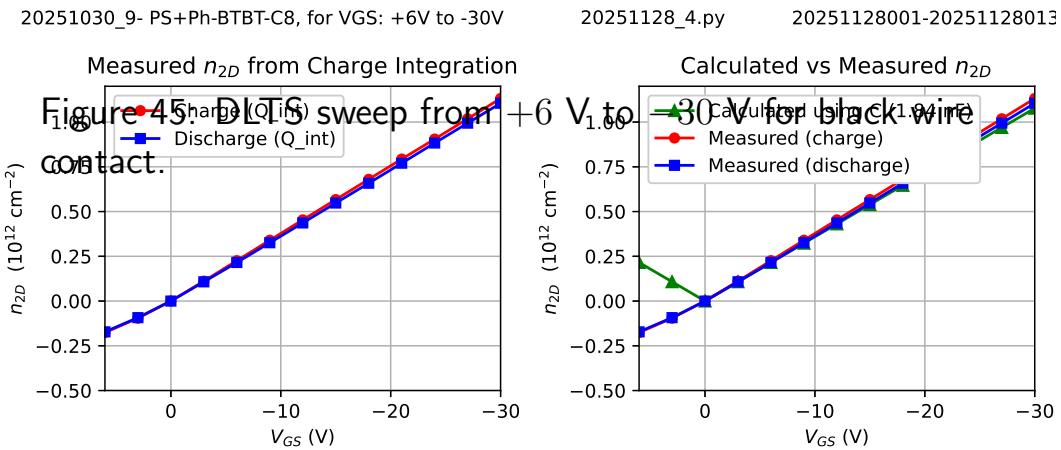



Figure 45: DLTS sweep from $+6$ V to $-30$ V for black wire contact.

## Gated Vander Pauw Measurements

**Measurement details:** Gated van der Pauw measurements on sample 20251030_9_PS+Ph-BTBT-C8 for source currents from 100 nA to 500 nA, with gate voltage from $-5$ V to $-30$ V.

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251125/20251125004`
- File numbers: 100 nA: `20251125004_01.dat`, 200 nA: `20251125004_02.dat`, 300 nA: `20251125004_03.dat`, 400 nA: `20251125004_04.dat`, 500 nA: `20251125004_05.dat`
- Python: `20251125004.py`
- EPS: `yySec010046.eps`

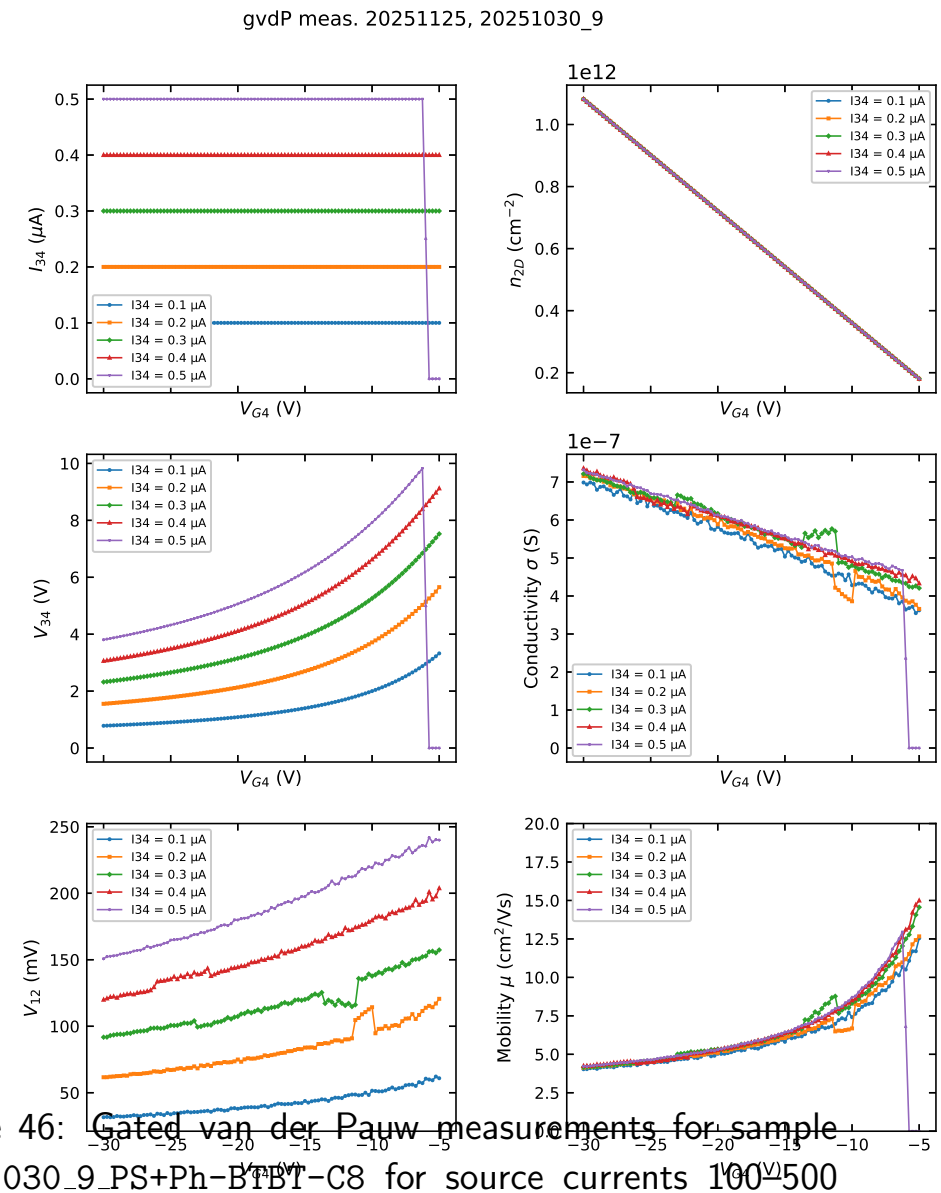



Figure 46: Gated van der Pauw measurements for sample 20251030_9_PS+Ph-BTBT-C8 for source currents 100–500 nA.

## IDS vs VDS Measurements

**Measurement details:** IDS vs VDS characteristics measured sampple 20251030_9_PS+Ph-BTBT-C8 $V_{GS} = 0, -5, -10, -15, -20, -25, +5$ V

- Data directory: `Q:/THz-Lab/Measurement Data/AllMeasuredData/2025/20251203/IDS vs VDS`
- File numbers: `20251203001.dat` – `20251203007.dat`
- Python: `20251203.py`
- EPS: `yySec010047.eps`

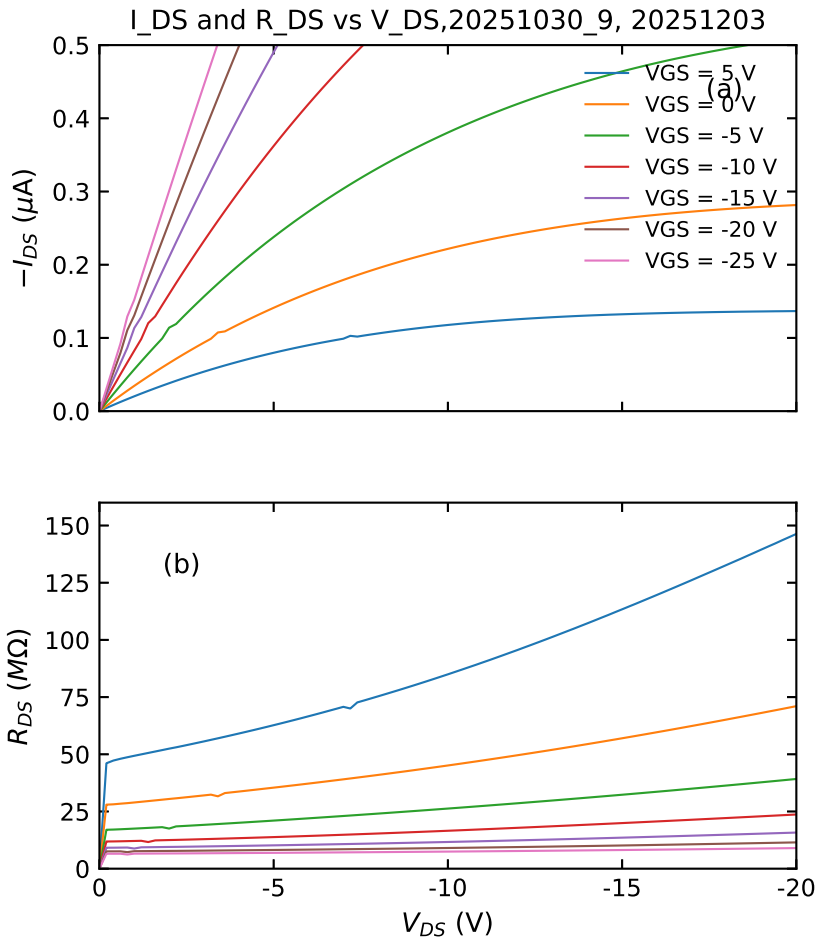



Figure 47: IDS vs VDS measurements for $V_{GS} = 0$ to $-25$ V and $+5$ V.

# 11 Starting the Mai Tai laser

**Date of issue/modification:** 14.01.2026

**Last Changes:** 14.01.2026

## 11.1 General remarks

1. Obey laser safety rules!
2. The laser is designed for long-time operation.
3. The laser will be damaged when
   - switching the laser on and off every day
   - drawing more than 4.8 W green power
4. In case of malfunction, the green power will be smaller than the desired value. In this case the laser automatically increases the diode's current. Such an increased current can kill the diodes within minutes. The costs for new diodes are about 25 000 EUR.
5. Operation in `Control Mode Green Power` is not recommended, because of the above.
   A remedy is to run the laser in `IR Power`. Then the output power is held constant.

## 11.2 Serial number and other numbers

1. in the back of mainframe
   - MFG P/N: J40-8S-26K-08
   - Cust P/N: PS-J40-8S-26K-08
   - December 2003
   - Serial number: 20568

## 11.3 Technical support

- 06151 708-0, Darmstadt 64291, Guerickeweg 7
- Herr Wang, +49-6151-708-0,+49-151-53032484
  fangzhou.wang@mksinst.com

## 11.4 Starting the chiller

1. Device: Thorlabs LK220
2. The device to be started first is the chiller.
3. Start the chiller before you turn any knob at the laser.
4. Check for sufficiant cooling liquid in the chiller.
5. If needed, refil with Coolflow DTX from Thorlabs.
6. Switch on the chiller.
7. The pump works only if the display reads RUN.
   If not, change from STOP to RUN.
8. The desired temperature is 20.0 celsius.
9. Check that the pump drive is set to 50%.
10. The flow meter should read 0.5 LPM.
11. Check that the bead of the flowmeter reacts to changes of the flowmeter's knob.
12. Set the flow to 0.5 LPM.
13. Wait until the temperature is stable at 20 celsius.

## 11.5 Starting the main frame of the laser

As of 08.12.2025:

1. Turn the key to ON.
2. Switch the orange 0/1 button to ON.
3. The display lights up:
   - Tests will be run automatically; wait until completed.
   - Then the main frame will warm up and also the SHG unit.
   - Wait until the display shows that 100% are reached.

## 11.6 Starting the computer

As of 08.12.2025:

1. The PC's name is: `LSF-WS-Treene`.
2. The MAC is: `00 19 99 6C AE D1`
3. The user ID is: `lsf-ws-treene\admin`
4. The password is: `tera-hertz!!`

## 11.7 Starting the program A before 2026

As of 08.12.2025:

1. Don't start the program before the Mai Tai has warmed up.
2. On the PC go to the directory:
   `Computer/Data(E:)\Mai Tai Laser Software ... Maitai_FinalTest_rev_L\`
3. Start the program: `Final.Test_rev_L.exe`
4. The proram will find the laser at COM-port 2.
5. The next panel asks where to load the calibration data from.
   Select `from the Mai Tai`.

## 11.8 Starting laser emission with program A

As of 12.02.2026:

1. Verify that the main frame display indicates 100%.
2. To initiate laser emission, press and hold the ON/OFF button in the GUI of the program until the laser starts.
3. Switch to Green Power mode and slightly adjust the wavelength above or below 780 nm. The program will automatically return to the set wavelength value.
4. Click on PZT Controls to open the corresponding window and monitor the IR Power Loop. The Green Correction bar should be green and approximately at a value of 50 to ensure stable operation.
5. Click on Info section and monitor the current and temperature of Diode 1.

6. If the Green Power remains one step below the desired value listed in the table below, adjust the Green Power by $\pm 0.1$ W to increase the diode current.
7. **The target operating values in Pump Mode are:**

| Wavelength (nm) | Green power (W) | Mill. current (%) | IR output (W) |
|---|---|---|---|
| 775 | 4.1 | 88.7 | 0.69 |
| 780 | 4.1 | 88.7 | 0.69 |
| 785 | 4.1 | 88.8 | 0.69 |

8. Once these values are reached, switch the laser to IR Power mode to stabilize the output power.
9. If the desired power level is achieved but pulsing is not observed in the operation window, open the Calibration window. In the RF Phase section, modify the New Value parameter by increasing or decreasing it relative to the value of 2 until the pulsing indicator in the main window turns green.
10. To turn off the laser and terminate operation, press the button labeled **DON'T SAVE, JUST EXIT**.

## 11.9 Starting the program B after 2025

As of 14.01.2026:

1. Do not start the program before the Mai Tai laser has completed its warm-up phase.
2. The program folder is located on the PC desktop under the name: 0077-1.03.09 MaiTi Customer GUI.
3. Launch the software by double-clicking the Mai Tai program icon on the desktop.
4. The program will automatically detect the laser at COM-port 2.

## 11.10 Starting laser emission with program B

As of 14.01.2026:

1. Verify that the main frame display indicates 100%.
2. Set the chiller working point to 20.0 °C at 50%.
3. Set the current limit of Diode 1 to 36.0 A.
4. To initiate laser emission, press and hold the ON/OFF button in the GUI of the program until the laser starts.
5. Click on View to open the Setup window.
6. Click on View to open the Info window.
7. **The target operating values in Pump Mode are:**

| Wavelength (nm) | Green power (W) | Mill. current (%) | IR output (W) |
|---|---|---|---|
| 775 | 4.1 | 88.7 | 0.69 |
| 780 | 4.1 | 88.7 | 0.69 |
| 785 | 4.1 | 88.8 | 0.69 |

8. Once these values are reached, switch the laser to IR Power mode to stabilize the output power.
9. If the desired power level is reached but pulsing is not observed in the RF section of the Setup window, perform

one of the following actions:

- Toggle the Modelocker enable switch off and then on, or
- Slightly adjust the wavelength above or below 780 nm, or
- Modify the RF Phase Percent value by increasing or decreasing it relative to the value of 2.